DKM

COMPLETE TECHNICAL HANDBOOK

Architecture, source-file atlas, algorithms, contract flows, security boundaries and verification evidence

Decentralized Knowledge Marketplace | Local college demonstration | Polygon Amoy

Snapshot date: 23 October 2026

293 repository files inventoried; 45 HTTP route declarations extracted; 19 database table names identified from schema creation code.

Prepared from the current working tree, retained experiment artifacts and recorded engineering evidence. This is a technical explanation and source map, not a claim that all code has been independently audited or all features executed live.

How to use this handbook

- Read Chapters 1-12 for the complete system story and evidence boundaries.

- Use the route, schema and configuration catalogs to trace data across layers.

- Use the file atlas for each file's role, code entry points with line numbers, dependencies and test cases.

- Use the code excerpts and runbook to connect conceptual explanations to the implementation.

Credentials, private keys, environment values, database dumps, installed third-party package source, generated build/cache trees and transient logs are not reproduced. Source line numbers refer to this snapshot. Environment variable names are documented without values.

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Practical guide A. Follow one asset end to end

Prepared 23 October 2026 from the current working tree. This guide expands the source-indexed handbook with operational explanations. Recorded test results remain historical evidence; preparing this PDF did not execute purchases, auctions or a new full test suite.

Stage	Who acts and what changes
1. Account	The browser signs up/logs in through FastAPI. PostgreSQL stores the account and hashed session records. A wallet-signature challenge binds the selected wallet to the account.
2. Analyze	The backend extracts a bounded text excerpt, asks Ollama for structured metadata and evidence-backed rubric scores, and embeds the generated summary with MiniLM.
3. Store	BLAKE3 identifies the plaintext bytes. AES-GCM encrypts them. Pinata pins encrypted content and metadata; PostgreSQL stores the catalog, vectors, CIDs and wrapped content key.
4. Mint	MetaMask signs an Amoy transaction to the registry. A successful receipt creates the registration token. The upload record is linked to the verified token; upload alone is not minting.
5. List	The seller submits a separate listing transaction. The marketplace stores the price and seller. Cancel listing closes availability; it does not remove the IPFS file.
6. Discover	Search embeds the query, fetches published candidates by cosine similarity, removes duplicate identities and applies MMR. The result opens the asset page.
7. Review and buy	The backend makes an expiring quote. The buyer accepts, approves MockUSDC if needed, and signs buyAsset. The contract transfers tokens, splits proceeds and records access atomically.
8. Access and download	The backend verifies the session, wallet proof, active asset and creator/purchase entitlement. It fetches matching ciphertext, unwraps the key, decrypts and verifies the plaintext hash before returning a download.

There is no single atomic transaction spanning PostgreSQL, IPFS and Amoy. The payment transaction is atomic on-chain, but uploading, minting, linking and downloading are separate operations. Recovery bridges interrupted stages. A successful upload with a failed wallet transaction is a resumable draft, not a purchased or minted asset.

What blockchain does - and does not do

Polygon Amoy is an existing public test network, chain ID 80002. DKM deploys Solidity contracts to it; DKM does not operate its own consensus network or mine POL. MetaMask holds the user signing authority, the RPC transports requests and contract reads, and validators execute transactions. CIDs and registration/access state are on-chain; full documents, login accounts, embedding vectors and master keys are not.

An ERC-721 registration identifies the asset. This marketplace sells access: buyAsset and endAuction update purchaseCount; they do not transfer the registration NFT to the buyer. Registration also does not prove legal authorship. Source comments mentioning an NFT transfer must not override the executable implementation.

When something fails

- Upload succeeds, mint fails: resume the saved upload; recovery searches for a matching already completed mint before retrying. No secret recovery code is required.

- Wallet rejected: no signed transaction is submitted. A submitted transaction hash is different: check its receipt before retrying.

- On-chain purchase reverts: token transfers and entitlement changes revert together, although gas for an included failed transaction can still be spent.

- Purchase succeeds, gateway fails: access remains on-chain; retry download rather than buying again.

- Database or master key lost: an on-chain purchase cannot reconstruct encrypted files or wrapped-key records. Preserve database, master key and pinned content.

Source anchors: backend/main.py; services/mint_recovery.py, blockchain.py, auth.py; frontend/app/mint/page.tsx; contracts/contracts/Marketplace.sol.

Practical guide B. Find and inspect the database

Docker inspection on 23 October 2026 found container dkm-postgres running image pgvector/pgvector:pg15, with PostgreSQL published on host port 5432. This establishes the running container and port, not a fresh audit of its data or backup. The backend selects its database through DATABASE_URL. The code fallback names database dkm; verify your local configured target before connecting.

Location	What lives there
PostgreSQL / pgvector	Accounts, sessions, catalog rows, 384D vectors, wrapped keys, quotes and retained workflow/history records.
Docker storage	PostgreSQL physical files live in the container data directory and any mounted volume/bind mount. Inspect mounts to find the actual persistence location.
Pinata / IPFS	Encrypted file bytes and public metadata objects. A CID is a content identifier, not a SQL row or a decryption key.
Polygon Amoy	Contract code/state, listings, auctions, token balances, allowance, access counters and transaction receipts/events.
Backend private configuration	Database connection details, master encryption key, provider credentials and RPC configuration. Never include values in a report.
research_results/	Saved QAOA/VQC metrics and reports; these are experiment artifacts, not live blockchain records.

Option 1: inspect through Docker Desktop

- Open Docker Desktop, select Containers, then dkm-postgres.

- Open its Exec/Terminal view. Use psql -U postgres -d dkm if those match your configured database role/name.

- If authentication is requested, use your existing local database credential. Do not reset the database or share the password.

- Use the read-only commands below. Exit psql with \q.

Option 2: PowerShell

```
docker ps --format "{{.Names}} | {{.Image}} | {{.Ports}}"
docker inspect dkm-postgres --format "{{json .Mounts}}"
docker exec -it dkm-postgres psql -U postgres -d dkm
```

The mount-only inspect command reports storage paths without dumping container environment variables. A Docker-managed volume normally lives in Docker Desktop's Linux storage, not as a directly editable Windows project folder. Use PostgreSQL tools rather than changing physical database files.

Read-only SQL you can use in psql

```
\conninfo
\dt public.*
\d knowledge_assets
\d app_users
BEGIN READ ONLY;
SELECT current_database(), current_user;
SELECT id, username, wallet_address, created_at
FROM app_users ORDER BY id DESC LIMIT 20;
SELECT id, title, token_id, price_usdc, quality, created_at
FROM knowledge_assets ORDER BY id DESC LIMIT 20;
SELECT COUNT(*) AS total_uploads,
       COUNT(token_id) AS linked_uploads
FROM knowledge_assets;
SELECT id, user_id, created_at, expires_at, accepted_at
FROM purchase_quotes ORDER BY created_at DESC LIMIT 20;
COMMIT;
\q
```

These queries avoid passwords, session hashes and wrapped keys. Account names/wallets are still personal data: inspect locally, not in public screenshots. A null token_id generally identifies an upload not linked to a mint. A catalog row is not proof of buyer entitlement; read the marketplace contract for that.

Option 3: graphical database client

In pgAdmin or DBeaver, add a PostgreSQL connection to localhost:5432. Enter the database, username and password matching your local DATABASE_URL privately. Browse public > Tables; choose knowledge_assets or app_users and use a limited read-only query. The pgvector extension provides the VECTOR(384) type inside PostgreSQL; it is not a separate vector database server.

Table map and safe operation

app_users holds account identity and password hashes/Google linkage; app_sessions holds hashed session tokens and expiry; auth_attempts tracks throttling; google_auth_flows holds temporary OAuth state. knowledge_assets connects content, vectors and chain identity; reviews stores review records; purchase_quotes stores accepted terms and expiry. Retained payment/provider/indexing tables support optional or historical workflows. Their presence does not mean those payment routes are active. The complete schema-source inventory appears later.

Do not use DELETE, TRUNCATE, docker volume rm or database recreation as a troubleshooting shortcut. A backup of PostgreSQL alone cannot restore decryption without the original master key. This document deliberately does not reproduce private configuration, keys or actual user rows.

Practical guide C. Quotes, money, verification and waiting

Three different prices

The seller chooses the asset/listing price. Ollama assessment is advisory and does not set the seller's final price. QAOA studies synthetic candidate prices offline and never changes a live listing. A purchase quote is a snapshot of an existing listing and fee distribution, not an exchange-rate service or an auction bid.

What you need in your Amoy wallet

Two distinct tokens are involved: MockUSDC pays the listing price; native test POL pays gas. A wallet containing 1 POL and zero MockUSDC cannot buy a 0.01 MockUSDC listing. MetaMask's displayed dollar estimate for POL does not convert it into MockUSDC. The current route uses test assets, not real dollar settlement. No INR/UPI/card conversion runs in the current wallet-only frontend.

Quote lifecycle

backend/services/payments/quotes.py creates purchase_quotes with expires_at = NOW() + 5 minutes. Terms bind the account's verified buyer wallet, chain, registry, marketplace, token/listing IDs, seller, payment token, exact price and fee split. Acceptance locks the quote row, checks ownership/expiry and rereads current chain terms. Changed terms or an expired quote require a new review. Acceptance neither transfers money nor grants access.

The quote is an application-level guard; buyAsset does not receive a quote ID or its expiry timestamp. The frontend validates before signing, but a transaction already submitted can be mined later. Do not describe quote expiry as an on-chain transaction cancellation. There is no reservation window: another buyer can purchase or the seller can cancel before your transaction executes.

Fee arithmetic: no extra DKM buyer fee

```
P = listing price in integer token base units
F = floor(P * platformFeeBps / 10000)
R_i = floor((P - F) * recipientShareBps_i / 10000)
Seller receives P - F - sum(R_i)
Buyer transfers P, plus separately paid POL gas.
```

With six decimals, 0.01 MockUSDC is 10,000 units. At the recorded 500-bps (5%) platform rate and zero royalties: treasury gets 500 units (0.0005), seller gets 9,500 (0.0095), buyer pays 0.01 plus gas. Configured royalties are calculated on the post-platform remainder. Quotes read the deployed splitter rate rather than relying only on this example. A payout-currency preference does not perform conversion.

What proves that payment happened

- The frontend reads balanceOf and allowance. approve authorizes spending; it is not a purchase.

- buyAsset checks listing availability, seller identity/ownership and sale type. It closes the listing and increments purchaseCount before external token calls.

- safeTransferFrom moves the payment token to RoyaltySplitter; the splitter distributes funds. If a transfer/distribution fails, all state changes in that transaction roll back.

- PurchaseCompleted is emitted on success. confirmedReceipt waits for two confirmations, requires a successful receipt and checks canonical block hashes and the current receipt.

- The backend verifies on-chain entitlement for protected access. It does not trust a UI success message, a submitted hash, a quote acceptance or an external-provider confirmation as entitlement.

Stage	How long / what ends it
Quote review	Five minutes from creation. Expiry requires another quote; funds are not locked.
Wallet confirmation	User-controlled. Nothing is guaranteed until the user signs and a transaction is submitted.
Approval transaction	Network-dependent, then two confirmations through the shared receipt helper. This does not buy the asset.
Purchase transaction	Network-dependent, then two confirmations and canonical-receipt checks. No fixed seconds guarantee is coded in the helper.
Pending RPC/wallet state	A timeout or lost browser response does not prove failure. Inspect the hash and entitlement before resubmitting.
Auction escrow	Winning bid stays in the contract until endAuction succeeds after endTime. Time passing alone does not execute it.
Outbid funds	Credited to pendingRefunds; remain there until the bidder calls withdrawRefund successfully.

For gas, the shared helper estimates the exact call, adds 20%, checks the wallet against gas * maxFeePerGas and respects the block/RPC cap. It uses the Amoy fee preparation with the observed 25-gwei tip floor. Reducing the tip below the node minimum causes rejection; increasing a gas limit to an enormous number does not make a transaction safer or cheaper.

Source anchors: quotes.py; frontend/lib/api/quotes.ts; frontend/lib/blockchain/purchaseSafety.ts; contracts/contracts/Marketplace.sol and RoyaltySplitter.sol. Optional provider implementations remain in source, but their confirmations must be separated from on-chain settlement and entitlement.

Practical guide D. Auction lifecycle and access release

An auction sells an access entitlement under the current implementation, not ownership transfer of the registration NFT. Inspect the actual contract logic rather than treating the word NFT in a UI or comment as proof of token transfer.

- createAuction records token, seller, positive reserve, startTime, endTime and royalty recipients. endTime must exceed startTime; the caller must own the registration and another auction must not already exist for the token.

- During startTime <= block.timestamp < endTime, a bidder other than the seller can placeBid. The first bid must meet reserve; a later bid must exceed the current highest bid.

- Each new bid transfers the full bid amount from that bidder into Marketplace. Even a repeated bid by the same wallet is a fresh transfer; the previous highest amount becomes a refund credit.

- The former highest bid is added to pendingRefunds instead of being pushed to the wallet during bidding. This prevents a failed outgoing refund from blocking every new bid.

- withdrawRefund sets the credit to zero before transferring it. A failed transfer reverts the state; a successful withdrawal emits RefundWithdrawn.

- After endTime, someone must submit endAuction. The contract marks it inactive/finalized and clears tokenToAuction.

- With no bids, it closes without payment or entitlement. With a winning bid, it increments the winner's purchaseCount, sends the escrowed tokens to the splitter and distributes proceeds. Failed payout reverts finalization and entitlement together.

Important operational boundaries

There is no general cancelAuction function in the inspected Marketplace. Do not present cancelListing as auction cancellation. endAuction is permissionless but requires a transaction and gas; it is not an automatic scheduled background action. It is paused by whenNotPaused, whereas withdrawRefund and cancelListing remain available while paused. The chapter describes source behavior, not a newly completed live auction demonstration.

How download authorization works after either purchase path

The same protected download policy follows fixed-price buying or a winning auction: authenticate account, verify wallet binding, read asset state and current/eligible legacy market entitlements, find the protected catalog row whose CIDs match the registration, then fetch ciphertext. The current/legacy distinction is bounded to the preserved original IDs 1-6, preventing a later legacy token with a reused number from granting access to a successor token.

AES-GCM first unwraps the per-file key using the backend master key and expected hash as AAD. It then decrypts the file using its nonce and the same AAD. Authentication failure rejects the operation. BLAKE3 of recovered plaintext must equal the expected stored digest. Only then is the file returned as a no-store attachment. Backend key custody means the operator can decrypt; the design is not end-to-end encrypted against the server.

Practical guide E. BLAKE3, AI and quantum terminology

BLAKE3 is the correct name

The requested black hashing is BLAKE3. quantum_hash(data) calls blake3.blake3(data).hexdigest(), a classical cryptographic hash. Identical bytes produce the same digest; changed bytes are expected to produce a different digest. This supports exact duplicate checks and integrity comparison, not paraphrase detection, plagiarism assessment, truth verification or a quantum algorithm.

The plaintext digest and encrypted IPFS CID identify different bytes. Random per-file keys/nonces mean encrypting the same plaintext again can produce different ciphertext and a different CID even though its BLAKE3 plaintext digest matches. Hashing is not encryption: the digest cannot recover the document. The key wrapper is stored in PostgreSQL, while CONTENT_MASTER_KEY remains backend configuration.

Term	Meaning and actual DKM role
VQA	Variational quantum algorithm: a family combining parameterized circuits and classical optimization. It is not a third independent production component here. QAOA and the trained VQC are examples of this approach.
QAOA (possibly your QA)	Quantum Approximate Optimization Algorithm. Implemented as a separate synthetic pricing experiment, executed on a classical simulator. Not called to set live asset prices.
VQC (sometimes called BQC)	Variational quantum circuit. Implemented as a four-qubit embedding experiment with training and classical reconstruction. Not used for production search embeddings.
QA as quality assessment	If QA means quality assessment, this is the active Ollama rubric workflow. It is classical LLM inference, not QAOA or quantum computing.
Legacy quantum search helper	Historical graph/QUBO/annealing helpers remain in source. Current production retrieval uses deterministic cosine/MMR, not these experiments.

What actually runs when the user searches

MiniLM produces normalized 384-dimensional embeddings for summaries and queries. PostgreSQL/pgvector retrieves cosine candidates. Published identity filtering and deduplication remove drafts/repeated items. MMR selects up to ten results by balancing relevance (0.85) and redundancy (0.15). Ollama produces metadata and criterion evidence during assessment; MiniLM, not Ollama, supplies these search vectors.

What was simulated and what the results mean

VQC: 96 authored documents, 32 queries (16 validation/16 test), four-qubit statevector circuit, two RY/RZ + CNOT layers, 16 trainable parameters and 40 epochs. Original MiniLM test NDCG@10 = 0.905296; PCA + ridge = 0.702724; untrained VQC + ridge = 0.605030; trained VQC + ridge = 0.609317. Training improves the VQC only slightly and does not beat the classical baselines. It is not 90.5% accuracy, and no quantum advantage or statistical significance is established.

QAOA: 64 synthetic demand cases, 100 candidate prices represented by seven qubits, depth two, up to 80 COBYLA evaluations and 2,048 seeded samples per case. Exact-price matches: 1/64 for QAOA and 64/64 for classical optimization. Mean per-case relative revenue gap = 17.6013%; mean feasible probability = 97.5540%. Mean times: QAOA 0.456721 s, classical 0.000505 s, exhaustive oracle 0.000051 s. These are simulated expected-revenue objectives, not real sales or hardware speedups.

The implementation and retained result files are genuine research components; they are not required for a buyer to search, pay or download. Their useful research conclusion is that this tested quantum representation/optimizer did not improve these small classical baselines. CPU simulation also does not establish that every embedding-preparation step ran on CPU; original benchmark hardware provenance should be documented separately.

How to explain the project in a viva

DKM combines authenticated accounts, a semantic catalog, encrypted IPFS storage and blockchain-recorded access. Wallet transactions enforce token settlement and access counters. The backend remains trusted for accounts, indexing, key custody and delivery. QAOA/VQC provide separately measured research comparisons. Claim each result at its evidence level: implemented code, automated test, simulation, receipt or complete workflow; do not substitute one for another.

1. System purpose and architectural boundaries

DKM connects user accounts, wallet-verified identity, knowledge-document registration, searchable descriptions and licensed access to encrypted content. It runs a local Next.js frontend and FastAPI backend, with PostgreSQL/pgvector and Ollama services, Pinata-hosted IPFS data, and public Polygon Amoy contracts.

```
Browser / MetaMask
  |-- Next.js pages, wallet transactions, query cache
  |-- FastAPI: sessions, metadata, uploads, search, access checks
        |-- PostgreSQL + pgvector: catalog, accounts, vectors, wrapped keys
        |-- Ollama: structured excerpt assessment / descriptions
        |-- MiniLM: normalized 384D embeddings
        |-- Pinata / IPFS: encrypted document + public metadata
        |-- Amoy RPC: registry, market, entitlements, receipts

Separate research path: PennyLane simulator -> QAOA/VQC results -> /research
```

A blockchain makes relevant contract state inspectable; it does not decentralize the local application server, its database or its master decryption key. An NFT registration is not proof of legal copyright, truth or scientific quality. In the current marketplace, a purchase grants access entitlement while the creator retains NFT ownership.

Layer	Responsibility	Trust / failure boundary
Frontend	Input, catalog display, wallet signing	Browser state is untrusted; UI hiding is not authorization
Backend	Validation, identity, storage, access and orchestration	Trusted with plaintext and decryption material
Database	Account/catalog/workflow state and vectors	Must preserve wrapped keys, ownership links and backups
IPFS / Pinata	Content-addressed encrypted storage	Pinning/gateway availability is not guaranteed permanence
Amoy contracts	Registration, settlement and entitlement state	Testnet gas/tokens; deployed code differs from local source
Research	Controlled comparative experiments	Simulated circuits, synthetic/authored data; no advantage claim

Active versus retained scope

The current customer UI is wallet-only. Transak, Razorpay, payout and receipt-recorder modules remain in the repository for historical work and optional future integration. Their existence does not mean those provider routes are enabled, funded, live or production-ready. Older TRNSK demo documents are superseded for the active MockUSDC purchase path.

2. Technology stack and why each part exists

Technology	Use in DKM
Python / FastAPI / Uvicorn	HTTP APIs, typed validation, async I/O and service lifecycle
Pydantic / httpx / asyncpg	Boundary schemas, outbound HTTP and PostgreSQL access
PostgreSQL + pgvector	Durable relational state and cosine vector candidate search
Next.js 14.2.35 / React 18 / TypeScript	Page routing and typed interactive frontend
Tailwind / reusable UI primitives / Lucide	Styling, accessible interaction primitives and icons
Wagmi / RainbowKit / viem	Wallet connection, chain-aware reads/writes, gas and receipt handling
Solidity / OpenZeppelin / Hardhat	Contract implementation, standard tokens/access guards and local tests
Ethers / Web3.py	Deployment/operational scripts and backend RPC reads
BLAKE3	Plaintext fingerprint and exact-byte integrity checks
cryptography AESGCM	Authenticated content encryption and key wrapping
Pinata / IPFS	Storage upload and retrieval of content-addressed objects
Ollama + OpenAI-compatible SDK	Local structured metadata and rubric scoring; SDK use does not imply OpenAI-hosted inference
Sentence Transformers / MiniLM	384D normalized semantic embeddings
NumPy / SciPy / scikit-learn	Vector math, optimization, PCA and ridge reconstruction
PennyLane	QAOA and VQC circuit simulation for experiments
NetworkX / dimod	Retained legacy graph/annealing helpers, inactive in production retrieval

Version evidence has different meanings: requirements files pin intended backend dependencies, package manifests may contain ranges, lockfiles resolve Node dependencies, and experiment JSON records the environment of a particular run. For example, current backend requirements pin NumPy 2.4.6 while retained VQC evidence records NumPy 2.5.1. Do not collapse these into one asserted runtime version.

3. Accounts, signup, login and wallet identity

The welcome page leads to login/signup. AppShell waits for the session before mounting protected application pages, and redirects authenticated users to the dashboard. API authorization is enforced independently on the backend. Signup data is stored in PostgreSQL app_users; if PostgreSQL runs in Docker, Docker hosts the database service, while persistence depends on its configured volume. Blockchain and Pinata are not the signup database.

Passwords use salted scrypt (n=2^17, r=8, p=1). The stored value contains a salt and derived key, not the plaintext password. A semaphore limits concurrent expensive password hashing per worker. Session cookies carry tokens whose SHA-256 digests are stored in app_sessions with expiry. HTTPS selects secure __Host- cookies; localhost uses the development cookie name. Exact Origin and X-DKM-Request checks protect state-changing account requests.

Wallet connection and wallet verification are different. MetaMask connection exposes an address; the backend challenge/signature flow proves control and binds a wallet to the logged-in account. Expiry and consumed challenges mitigate replay. Account-scoped uploads, recovery and reviews must not accept another user's identifiers simply because the browser submitted them.

Google OAuth plumbing exists, including state/PKCE-related persisted flows. Its availability depends on correct provider configuration; the historical checks recorded setup pending. This handbook does not certify a successful live Google login.

4. Upload, analysis, storage and NFT minting

- 1. The authenticated user chooses a supported document and requests analysis.

- 2. The backend reads bounded bytes and extracts an excerpt. It calls Ollama for descriptions and the rubric.

- 3. Upload re-analyzes the actual bytes server-side; browser-supplied quality cannot replace the server assessment.

- 4. The quality threshold is checked; the summary is embedded and exact duplicate/tiny-file checks run.

- 5. A BLAKE3 plaintext digest is computed, a random content key encrypts the file, and the master key wraps that content key.

- 6. Pinata receives encrypted content and public metadata. Their CIDs and protected catalog fields are persisted in PostgreSQL.

- 7. MetaMask signs KnowledgeAsset mint with CIDs, price, license and quality. This costs test POL gas, not the asset purchase price.

- 8. The frontend waits for a confirmed receipt, then the backend validates creator/CIDs/price and links the saved upload to the token.

The current minimum is 0.01 USDC = 10,000 base units of the six-decimal test token. Pricing is user-selected, not an LLM market valuation. Contract storage also carries license and active state; an inactive asset is not eligible for ordinary content access.

Why a file can exist on Pinata before minting succeeds

IPFS upload and wallet minting are separate operations. Rejecting or failing the wallet transaction does not undo an already completed upload. The dashboard preserves the draft and offers resume without requiring a mysterious code. A token ID exists only after a successful mint; manual linking is an advanced recovery option. Pending hashes are kept per upload/registry in browser storage. Matching completed mints are checked before another attempt, and backend recovery is account/wallet scoped. Automatic scanning is bounded to 100 tokens; cross-device concurrent requests are not a global exactly-once protocol.

5. BLAKE3 hashing and encryption in detail

quantum_hash(data) calls blake3.blake3(data).hexdigest(). With the default output it produces a 32-byte digest rendered as 64 hexadecimal characters. It is deterministic for the same bytes. Changing a file's formatting or metadata can change its digest even when the document looks similar. It is a classical cryptographic hash, not encryption and not a quantum circuit.

The digest is used for exact duplicate lookup, stored as blake3_hash, and bound into AES-GCM associated authenticated data. On protected download, the server decrypts and recomputes BLAKE3; a mismatch rejects the content. This does not detect paraphrased plagiarism, establish authorship or guarantee that uploaded scientific claims are true.

Envelope encryption

- A fresh 256-bit random data key is generated for each upload.

- A fresh 12-byte nonce encrypts the plaintext with AES-GCM. Stored ciphertext is nonce || encrypted bytes/tag.

- A separate random 12-byte nonce wraps the data key under the 32-byte backend master key. The wrapped envelope is base64-encoded for database storage.

- Both operations authenticate the ASCII content digest as associated data, binding ciphertext and wrapped key to the expected content identity.

- The master key is not stored in public IPFS metadata or frontend code. Losing it prevents decryption of protected content; compromising it threatens all wrapped keys accessible to the attacker.

AES-GCM provides authentication under correct nonce/key use. The application additionally checks the plaintext digest. Neither mechanism prevents an authorized recipient from redistributing downloaded plaintext. Older unencrypted IPFS assets remain publicly retrievable to anyone who has their CID; they are not retroactively made confidential.

6. IPFS, Pinata and the database

Pinata is the upload/pinning service used by backend/main.py through its files API. The encrypted document is stored as one object, public catalog metadata as another. IPFS addresses objects by content-derived identifiers; the encrypted-object CID is different from the plaintext BLAKE3 fingerprint. Encryption randomness means repeated encryption can produce a different CID for identical plaintext.

The database connects account ownership, vector ID, token ID, file CID, metadata CID, title/domain/tags/summary, quality, exact price, embedding, plaintext hash, original filename and wrapped content key. Public metadata must not contain the key or private assessment excerpts. The browser receives useful catalog fields rather than raw internal storage details.

On download, the backend checks current entitlement or creator access and active state, matches the on-chain CIDs to a protected database row, retrieves bounded ciphertext through configured gateways, decrypts and verifies it, and returns a no-store attachment with a sanitized filename. Gateway failures or authentication/integrity failures do not become successful downloads.

Backup boundary

The retained restore drill checked local PostgreSQL table counts and digests after restoration. It did not prove external IPFS recovery or independent master-key restoration. A complete recovery plan needs database + key backup + durable content availability + deployment references. Never publish any backup containing credentials or user records.

7. Ollama scoring and semantic retrieval

The default scoring model is llama3.2:3b via Ollama's OpenAI-compatible endpoint. Structured output uses temperature 0 and seed 31. Criteria are clarity, specificity, completeness and evidence, each an integer 0-4. Backend quality is their sum divided by 16; the acceptance threshold is 0.4. Positive criterion scores require supporting excerpt quotations, and invalid output is rejected rather than assigned invented quality.

Extraction is capped at 4,000 characters, with format-specific bounds such as five PDF pages or 51 CSV lines. Consequently, quality describes the available excerpt, not an exhaustive peer review. A quotation matching the excerpt does not prove that the assessment is justified. Two retained sanity examples scored 0.5625 (detailed, repeated) and 0.125 (vague); this is illustrative, not calibrated accuracy or deterministic behavior across all environments.

Embedding and search path

MiniLM is loaded lazily from the local cache. model.encode(..., normalize_embeddings=True) produces a 384D vector. Upload indexing uses the generated summary; query embedding uses the same encoder. pgvector stores VECTOR(384), with an IVFFlat cosine index and metadata filters. Candidate retrieval requests up to 60 published linked assets; deduplication excludes repeated content/token identities and drafts without useful token links.

```
For candidate d and already selected results S:
MMR(d) = 0.85 * cosine(query, d)
         - 0.15 * max(0, max cosine(d, s) for s in S)

Choose the largest MMR repeatedly, up to 10 results.
First result uses redundancy = 0.
Stable initial ordering uses descending query score and vector_id.
```

This is classical maximum marginal relevance. It reduces redundant results while retaining relevance; it is not a guarantee that every returned document is useful. The score is not a percentage probability. Legacy graph/quantum_walk_discover/QUBO helpers are retained but inactive in the production route. The old propagation is not unitary quantum evolution, and its annealing solver is classical.

8. Contracts, listings, purchases and auctions

Contract	Primary role
KnowledgeAsset	ERC-721 registration and mutable asset metadata/state within contract rules
KnowledgeAssetSuccessor	Read-through legacy prefix plus new registrations at a lower minimum
Marketplace	Listings, access purchases, funded settlement, auctions and refunds
RoyaltySplitter	Treasury fee, configured royalties and seller remainder
MockERC20	Unrestricted test-token mint and ERC-20 balances/allowances
PaymentReceiptRegistry	Optional audit receipt recording, not access entitlement

A listing checks ownership and the required approvals and records sale terms. Fixed-price buyAsset validates the listing is active and fixed-price, rejects a seller buying their own listing, and checks the seller still owns the NFT. It deactivates the listing and increments the buyer's purchase count, then transfers the ERC-20 price to RoyaltySplitter and distributes it. All effects revert if token transfer/distribution fails. PurchaseCompleted records the sale. The NFT is not transferred by this purchase path.

Auctions add reserve/start/end constraints, bids, outbid refund accounting, settlement and cancellation/finalization rules. Time-based eligibility requires block-time comparisons and careful expiry tests. Refund withdrawals are distinct transactions. A frontend button is not proof that any particular live auction has completed.

Current deployment and legacy compatibility

Component	Current Amoy address
KnowledgeAssetSuccessor	0xCA6f1E6C08Dddda08e4288B8eFE07bF54f67AeD7
Marketplace	0xc1cfA63fDA41cD1Ee76B24F1D1B2b993661a67A5
RoyaltySplitter	0x36CAa5b59825a378664E75b5EFB30988F71901c0
MockUSDC	0x0bd22C54F90E7309dd5e7e73270Ac72d4cD74719

Original registry: 0x2C16409A3db9880f9A70e71B5b131A17db533Fc9. Previous marketplace: 0xDe7bdacb8B3ff631dbcdb09C81c0de7421475A5d. Previous splitter: 0x43cE72CEd667bDe9252044451b07ecA1aec2D300. Existing IDs 1-6 are read from the original registry; new NFTs start at 7. Entitlement lookups only consult legacy markets for that fixed prefix, preventing future legacy ID collisions from granting access to a new-registry asset.

Source was compiled with Solidity 0.8.37, optimizer runs 200 and Shanghai target for the successor transition. Do not assume older deployments use the same compiler. contracts/cents-deployment.json records deployment receipts and verification; locally editing MIN_PRICE_USDC would not change an old deployed contract by itself.

9. Exact current payment flow and test funding

The active market is Polygon Amoy chain 80002. Test MockUSDC pays the asset price; test POL pays gas. Native POL and ERC-20 MockUSDC are distinct balances. A wallet's dollar display is not an ERC-20 balance and does not automatically convert test POL into test USDC. Minting/listing NFTs normally needs gas only, while buying additionally needs the price in MockUSDC.

- Read the chain-verified market profile and exact six-decimal token amount.

- Create and accept a fresh buyer/asset/listing/token-bound quote; a quote does not reserve a listing.

- Recheck listing eligibility and read balanceOf(buyer). Reject insufficient payment-token funds.

- Read allowance; if needed, approve the marketplace for the required price and wait for confirmation.

- Estimate the purchase call, check test POL gas balance, recheck terms/listing and submit buyAsset.

- Confirm the receipt and query entitlement before exposing protected download access.

Shared Amoy fee preparation enforces chain ID, uses a 25-gwei priority-fee floor, estimates the actual call, adds 20% gas margin and checks the balance against the maximum fee. It rejects estimates beyond the observed RPC/block bounds. Lowering a fee below the RPC minimum does not save a successful transaction; it causes rejection.

Fee example at 0.01 USDC

Item	Amount
Buyer asset price	0.010000 test MockUSDC
Extra DKM buyer fee	0
Platform deduction at 500 bps (5%)	0.000500 test MockUSDC from seller gross
Seller remainder with no royalties	0.009500 test MockUSDC
Wallet gas	Separate variable test POL cost

Configured royalty recipients, when present, change the seller remainder according to RoyaltySplitter. Integer base units and basis points avoid floating-point payment arithmetic. A seller payout preference does not itself execute FX.

Previously confirmed test-token funding

At block 50772211, mint(address,uint256) issued 2,000,000 base units to 0x9202BfF93f8A46f7D9BAe982CB1c021e34E8dc66. The confirmed balance at that block was 2.0 MockUSDC. Gas spent was 0.002247510007042198 test POL. Transaction: 0x321780e016a918f8fe48b789f9176c7c5a17f7e6e29390cb92589b957fa5ea20. This proves test funding, not a subsequent purchase or download; later balances can change.

Retained external-provider architecture

Retained Razorpay collection requires separate prefunded settlement; it is not INR-to-USDC conversion. Transak completion and optional receipt recording cannot independently grant access. MockPaymentProvider is a non-spending workflow simulator, unlike the actual test-chain MockERC20. Neither retained provider code nor a successful test-token mint proves real UPI/card checkout or arbitrary cryptocurrency conversion.

Persisted order states include CREATED, PENDING, PROVIDER_CONFIRMED, SETTLEMENT_PENDING, SETTLED, ENTITLEMENT_GRANTED and FAILED. Idempotency and canonical receipt checks protect settlement. These retained paths are separate from the active wallet-only demo.

10. QAOA and VQC: implemented experiments

VQC embedding experiment

The authored corpus has 96 documents across eight domains, with four topics/domain and three variants/topic. There are 32 queries split 16 validation/16 test; 64 document vectors fit preprocessing/model parameters and 32 variants are held out from fitting. Topics still occur in fitting documents, so this is in-domain retrieval. Labels 3/1/0 are authored rules rather than independent assessor judgments.

PCA reduces vectors to four components; arctan-scaled values encode into four qubits. Two layers of RY/RZ and ring CNOTs produce four Z and four adjacent ZZ expectations. Sixteen trainable parameters are optimized for 40 epochs using cosine-Gram alignment. Ridge expansion maps features back to 384 dimensions; it does not recreate discarded information. PennyLane executes exact classical statevector simulation.

Test NDCG@10 method	Result
Original MiniLM	0.905296
PCA4 + ridge expansion	0.702724
Untrained VQC + ridge	0.605030
Trained VQC + ridge	0.609317

Training took about 1.027 seconds in the retained run. Loss decreased from 0.485185 to 0.098681, but retrieval still underperformed the classical baselines. NDCG is a ranking metric, not classification accuracy. Single-seed, small-corpus results do not establish broad superiority or significance.

QAOA pricing experiment

Sixty-four synthetic demand cases use 100 integer price choices, seven qubits, depth two, COBYLA with up to 80 evaluations and 2,048 seeded samples/case. The cost Hamiltonian encodes negative normalized expected revenue with invalid-state penalties. The most frequent feasible sampled price is selected. Classical bounded optimization and exhaustive enumeration are comparators.

Metric	Result
Mean QAOA relative revenue gap	17.6013%
Mean feasible probability	97.5540%
Exact-price matches: QAOA / classical	1 of 64 / 64 of 64
Mean elapsed: QAOA / classical	0.456721 s / 0.000505 s
Mean elapsed: exhaustive oracle	0.000051 s

These are synthetic expected-revenue units, not actual sales. Small independent price problems favor classical optimization. Full statevector/diagonal simulation has exponential costs; no hardware quantum speedup or scalable oracle is established. Neither experiment changes production search vectors or live prices.

11. Tests, evidence and what has actually been proved

Evidence is separated into source implementation, local automated checks, simulation/read-only chain checks, confirmed public transactions and full user workflows. None implies all the others. This handbook does not rerun the entire suite.

Evidence	Supported conclusion	Not established
131 contract tests recorded after successor change	Local tested contract behavior at that snapshot	Every operation performed on Amoy
17 frontend tests + production build recorded	Covered regressions/build passed	All browser/wallet flows completed
3 recovery tests and 9 backend checks in separate runs	Targeted recovery/account/legacy/capability behavior	One combined fresh exhaustive test count
Six legacy cancel simulations	Calls estimated successfully at observed state	Actual cancellation receipt
Live one-cent mint estimate	Call was simulatable at given state	User NFT mint completion
Confirmed 2 MockUSDC mint	Token issuance and balance at block 50772211	Purchase entitlement or download
Retained VQC/QAOA JSON	Specific comparative experiment results	Quantum advantage / production retrieval quality
Local DB restore digest comparison	Restored tables matched captured data	IPFS/master-key disaster recovery
Saved Slither report	Recorded static findings at analyzed source	Independent production security audit

Earlier acceptance evidence recorded 82 backend tests, 15 frontend regressions and 129 contract tests at a previous snapshot. Newer targeted counts supersede only their stated scope. Do not add overlapping counts. Saved static analysis reported zero high/medium, four low timestamp findings and one informational finding; retain the analyzed revision and tool output when citing it.

Remaining final demonstration

- Use a distinct seller and buyer account/wallet. Confirm buyer has MockUSDC and POL on Amoy.

- Complete allowance approval and a real testnet purchase; save the successful receipt and verify purchaseCount/entitlement.

- Download as buyer and compare the plaintext hash with the original upload.

- Verify a third unrelated wallet is denied protected download.

- If claiming live auctions/cancellation, record those receipts separately; do not substitute a simulation.

- Document any failure and retry honestly, including gateway/RPC availability.

12. Contributions, limitations and security review

Project-specific engineering contributions include integrating encrypted content with blockchain-backed access, account/wallet-scoped recovery, a legacy-preserving one-cent registry transition, duplicate-aware semantic retrieval, excerpt-grounded rubric validation, exact payment-unit handling and transparent quantum/classical comparison artifacts. These are integrations and implementations, not inventions of BLAKE3, AES-GCM, MiniLM, MMR, QAOA or variational circuits.

- Implemented protections include session expiry/revocation, origin checks, signature verification, auth rate limits, encrypted content, transaction preflights, payment idempotency and account boundaries.

- The backend is a trusted plaintext/key custodian. A stolen master key and database access can compromise protected files.

- Public legacy files remain public. Decrypted content can be copied by authorized recipients.

- Google/provider availability depends on configuration and external services; old evidence can expire.

- Index backfill coverage is incomplete in retained evidence; do not assert complete lifetime totals.

- No independent audit, production load proof, calibrated quality accuracy or universal payment support is claimed.

- Post-quantum key release remains proposed, not implemented. The quantum_hash filename is not post-quantum protocol evidence.

13. API endpoint catalog

Extracted directly from FastAPI app/router decorators. Prefixes are combined for routers with literal prefixes. Line numbers identify the declaration in this snapshot. Authentication and state guards must be read in the handler/dependencies; a route name alone does not establish who may call it.

GET /

backend/main.py:134

GET /health

backend/main.py:139

POST /api/analyze

backend/main.py:195

POST /api/upload-ipfs

backend/main.py:218

POST /api/search

backend/main.py:417

GET /api/assets/{token_id}/access

backend/main.py:428

GET /api/assets/{token_id}/download

backend/main.py:452

GET /api/assets/{token_id}/details

backend/main.py:507

POST /api/assets/{vector_id}/recover-mint

backend/main.py:549

POST /api/assets/{vector_id}/link-token

backend/main.py:576

GET /api/account/dashboard

backend/main.py:601

GET /api/capabilities

backend/main.py:613

GET /api/account/reviews

backend/main.py:620

GET /api/assets/{token_id}/payment

backend/main.py:637

GET /api/account/activity

backend/main.py:648

POST /api/payments/webhook/razorpay

backend/main.py:681

POST /api/payments/{order_id}/settle

backend/main.py:720

POST /api/payments/create

backend/main.py:730

GET /api/payments/{order_id}

backend/main.py:753

POST /api/payments/{order_id}/confirm

backend/main.py:772

POST /api/auth/signup

backend/services/auth.py:141

POST /api/auth/login

backend/services/auth.py:158

GET /api/auth/methods

backend/services/auth.py:182

POST /api/auth/google/start

backend/services/auth.py:187

GET /api/auth/google/callback

backend/services/auth.py:209

GET /api/auth/me

backend/services/auth.py:260

POST /api/auth/logout

backend/services/auth.py:266

POST /api/auth/wallet/challenge

backend/services/auth.py:283

POST /api/auth/wallet/verify

backend/services/auth.py:294

GET /api/account/indexed-activity

backend/services/event_indexer.py:214

POST /api/purchase-quotes/asset/{token_id}

backend/services/payments/quotes.py:125

POST /api/purchase-quotes/{quote_id}/accept

backend/services/payments/quotes.py:134

GET /api/payments/recordings

backend/services/payments/recording.py:273

GET /api/payments/transak/status

backend/services/payments/transak.py:129

POST /api/payments/transak/session

backend/services/payments/transak.py:140

POST /api/payments/transak/orders/refresh

backend/services/payments/transak.py:247

GET /api/payments/transak/orders

backend/services/payments/transak.py:256

GET /api/payments/transak/payout-preference

backend/services/payments/transak.py:312

PUT /api/payments/transak/payout-preference

backend/services/payments/transak.py:323

POST /api/payments/transak/payout-session

backend/services/payments/transak.py:341

GET /api/payments/transak/coverage

backend/services/payments/transak_discovery.py:122

GET /api/account/positions

backend/services/positions.py:47

GET /api/research/results

backend/services/research_results.py:21

GET /api/assets/{token_id}/reviews

backend/services/reviews.py:16

POST /api/assets/{token_id}/reviews

backend/services/reviews.py:33

14. Database and configuration catalog

Table names below are extracted from CREATE TABLE IF NOT EXISTS statements, not from a live schema dump. Additional ALTER statements extend their columns. Source files are the authoritative place to inspect constraints and migrations.

app_sessions

Schema source: backend/services/auth.py

app_users

Schema source: backend/services/auth.py

asset_funding

Schema source: backend/services/payments/asset_funding.py

auth_attempts

Schema source: backend/services/auth.py

chain_sync

Schema source: backend/services/event_indexer.py

google_auth_flows

Schema source: backend/services/auth.py

knowledge_assets

Schema source: backend/services/pgvector_client.py

market_events

Schema source: backend/services/event_indexer.py

payment_evidence

Schema source: backend/services/payments/recording.py

payment_orders

Schema source: backend/services/pgvector_client.py

payment_recovery_events

Schema source: backend/services/pgvector_client.py

payout_observations

Schema source: backend/services/payments/payouts.py

payout_resolutions

Schema source: backend/services/payments/payouts.py

purchase_quotes

Schema source: backend/services/payments/quotes.py

receipt_jobs

Schema source: backend/services/payments/recording.py

reviews

Schema source: backend/services/pgvector_client.py

seller_payouts

Schema source: backend/services/payments/payouts.py

transak_staging_orders

Schema source: backend/services/payments/transak.py

transak_staging_tokens

Schema source: backend/services/payments/transak.py

Environment variable names only

Values and defaults are intentionally withheld. These names are statically referenced in application/source files and may include inactive provider paths. Presence in this list does not mean configured, enabled or safe to expose. Server secrets must not receive NEXT_PUBLIC_ prefixes.

ALCHEMY_AMOY_URL

Referenced by: contracts/hardhat.config.ts, contracts/scripts/deploy-cents.mjs

BLOCKCHAIN_CHAIN_ID

Referenced by: backend/main.py, backend/services/blockchain.py

BLOCKCHAIN_RPC_URL

Referenced by: backend/check_demo_readiness.py, backend/main.py, backend/services/blockchain.py, backend/services/payments/capabilities.py, backend/services/payments/recording.py, backend/services/payments/settlement.py, backend/verify_transak_token.py

CONTENT_MASTER_KEY

Referenced by: backend/services/content_security.py

DATABASE_URL

Referenced by: backend/main.py, backend/services/pgvector_client.py

DKM_APPROVED_AMOY_DEPLOYMENT

Referenced by: contracts/scripts/deploy-trnsk-sandbox.ts

DKM_DEPLOY_CENTS

Referenced by: contracts/scripts/deploy-cents.mjs

EVENT_INDEXER_ENABLED

Referenced by: backend/main.py

FRONTEND_ORIGIN

Referenced by: backend/services/auth.py

KNOWLEDGE_ASSET_ADDRESS

Referenced by: backend/check_demo_readiness.py, backend/migration_inventory.py, backend/services/blockchain.py, backend/services/payments/capabilities.py, backend/services/payments/recording_sources.py

LEGACY_KNOWLEDGE_ASSET_ADDRESS

Referenced by: backend/services/blockchain.py

LEGACY_MARKETPLACE_ADDRESSES

Referenced by: backend/services/blockchain.py, backend/services/payments/capabilities.py

LEGACY_TOKEN_LIMIT

Referenced by: backend/services/blockchain.py

MARKETPLACE_ADDRESS

Referenced by: backend/check_demo_readiness.py, backend/services/blockchain.py, backend/services/payments/capabilities.py, backend/services/payments/settlement.py

NEXT_PUBLIC_BACKEND_URL

Referenced by: frontend/lib/api/client.ts

NEXT_PUBLIC_WC_PROJECT_ID

Referenced by: frontend/lib/blockchain/wagmi.ts

OLLAMA_BASE_URL

Referenced by: backend/services/llm_enrichment.py

OLLAMA_MODEL

Referenced by: backend/services/llm_enrichment.py

PAYMENT_RECEIPT_REGISTRY_ADDRESS

Referenced by: backend/services/payments/recording.py

PINATA_JWT

Referenced by: backend/main.py

PRIVATE_KEY

Referenced by: contracts/hardhat.config.ts, contracts/scripts/deploy-cents.mjs

RAZORPAY_KEY_ID

Referenced by: backend/check_demo_readiness.py, backend/services/payments/razorpay_provider.py, backend/services/payments/recording_sources.py, backend/services/payments/service.py

RAZORPAY_KEY_SECRET

Referenced by: backend/main.py, backend/services/payments/razorpay_provider.py, backend/services/payments/recording_sources.py, backend/services/payments/service.py

RAZORPAY_TEST_INR_PER_USDC

Referenced by: backend/services/payments/razorpay_provider.py

RAZORPAY_WEBHOOK_SECRET

Referenced by: backend/main.py

RECEIPT_RECORDER_PRIVATE_KEY

Referenced by: backend/services/payments/recording.py

RECEIPT_RECORDING_ENABLED

Referenced by: backend/main.py, backend/services/payments/recording.py

RPC_LOG_PAGE_SIZE

Referenced by: backend/services/activity.py, backend/services/event_indexer.py

SETTLEMENT_ENABLED

Referenced by: backend/main.py, backend/services/payments/capabilities.py, backend/services/payments/settlement.py

SETTLEMENT_PRIVATE_KEY

Referenced by: backend/check_demo_readiness.py, backend/services/payments/capabilities.py, backend/services/payments/recording.py, backend/services/payments/settlement.py

TRANSAK_RECONCILIATION_ENABLED

Referenced by: backend/main.py

TRANSAK_STAGING_API_KEY

Referenced by: backend/services/payments/transak.py

TRANSAK_STAGING_API_SECRET

Referenced by: backend/services/payments/transak.py

15. Core source excerpts and how to read them

The following are actual selected implementation excerpts, not a replacement implementation. Long lines are wrapped for print. The file atlas supplies the broader symbol map, including all functions/classes identified in each source file. Secrets and environment values are not reproduced.

backend/services/quantum_hash.py :: quantum_hash

```
def quantum_hash(data: bytes) -> str:
    """
    Generate a BLAKE3 hash of binary data.
    """
    return blake3.blake3(data).hexdigest()
```

backend/services/quantum_hash.py :: verify_integrity

```
def verify_integrity(data: bytes, expected_hash: str) -> bool:
    """
    Verify that the supplied data matches the expected hash.
    """
    return quantum_hash(data) == expected_hash
```

backend/services/content_security.py :: seal

```
def seal(data: bytes, content_hash: str):
    master = master_key()
    key = AESGCM.generate_key(bit_length=256)
    nonce, wrap_nonce = os.urandom(12), os.urandom(12)
    aad = content_hash.encode("ascii")
    ciphertext = nonce + AESGCM(key).encrypt(nonce, data, aad)
    wrapped = wrap_nonce + AESGCM(master).encrypt(wrap_nonce, key, aad)
    return ciphertext, base64.b64encode(wrapped).decode("ascii")
```

backend/services/content_security.py :: unseal

```
def unseal(ciphertext: bytes, wrapped: str, content_hash: str):
    envelope = base64.b64decode(wrapped, validate=True)
    aad = content_hash.encode("ascii")
    key = AESGCM(master_key()).decrypt(envelope[:12], envelope[12:], aad)
    return AESGCM(key).decrypt(ciphertext[:12], ciphertext[12:], aad)
```

backend/services/embedding.py :: generate_embedding

```
def generate_embedding(text: str) -> list[float]:
    model = get_model()

    vector = model.encode(
        text,
        normalize_embeddings=True
    )

    return vector.tolist()
```

backend/services/quantum_search.py :: rank_results

```
def rank_results(candidates: list, top_k: int = 10) -> list:
    """Deduplicate identities, then use deterministic maximum marginal relevance."""
    unique, tokens, hashes = [], set(), set()
    for asset in sorted(candidates, key=lambda a: (-a["score"], a["vector_id"])):
        token, content_hash = asset.get("token_id"), asset.get("blake3_hash")
        if not token or asset["score"] <= 0 or token in tokens or (content_hash and
content_hash in hashes):
            continue
        tokens.add(token)
        if content_hash:
            hashes.add(content_hash)
        unique.append(asset)
    selected = []
    while unique and len(selected) < top_k:
        def marginal(asset):
            vector = np.asarray(asset["embedding"], dtype=float)
            redundancy = 0.0
            for previous in selected:
                other = np.asarray(previous["embedding"], dtype=float)
                denominator = np.linalg.norm(vector) * np.linalg.norm(other)
                similarity = float(vector @ other / denominator) if denominator else 0.0
                redundancy = max(redundancy, similarity)
            return 0.85 * asset["score"] - 0.15 * redundancy
        best = max(range(len(unique)), key=lambda i: marginal(unique[i]))
        selected.append(unique.pop(best))
    # Catalog output only: embeddings, storage references and fingerprints stay internal.
    fields = ("vector_id", "token_id", "title", "summary", "domain", "tags", "quality",
"price_usdc", "score")
    return [{key: asset.get(key) for key in fields} for asset in selected]
```

backend/services/auth.py :: password_hash

```
def password_hash(password, salt=None):
    salt = salt or secrets.token_hex(16)
    key = hashlib.scrypt(password.encode(), salt=bytes.fromhex(salt), n=2**17, r=8, p=1,
maxmem=256*1024*1024)
    return salt + ":" + key.hex()
```

backend/services/auth.py :: same_origin

```
def same_origin(request: Request):
    # Custom header forces a CORS preflight; exact Origin also rejects sibling sites.
    if request.headers.get("origin") != ORIGIN or request.headers.get("x-dkm-request") != "1":
        raise HTTPException(403, "Request origin is not allowed")
```

contracts/contracts/MockERC20.sol

```
function mint(
        address to,
        uint256 amount
    )
        external
    {
        _mint(to, amount);
    }
```

contracts/contracts/Marketplace.sol

```
function _buyAsset(uint256 listingId, address buyer) internal {
        Listing storage listing = listings[listingId];

        if (!listing.active)
            revert ListingInactive();

        if (listing.saleType != SaleType.FixedPrice)
            revert InvalidSaleType();

        if (listing.seller == buyer)
            revert SellerCannotBuyOwnAsset();

        if (knowledgeAsset.ownerOf(listing.tokenId) != listing.seller)
            revert SellerNoLongerOwnsNFT();

        uint256 price = listing.priceUSDC;
        address seller = listing.seller;
        uint256 tokenId = listing.tokenId;

        //--------------------------------------------------------------
        // Effects: close out the listing before any external call
        //--------------------------------------------------------------

        listing.active = false;

        delete tokenToListing[tokenId];

        // Entitlement rolls back with the transaction if payment/distribution fails.
        purchaseCount[tokenId][buyer] += 1;

        //--------------------------------------------------------------
        // Interactions
        //--------------------------------------------------------------

        usdc.safeTransferFrom(

            msg.sender,
            address(royaltySplitter),
            price
        );

        royaltySplitter.distributeRoyalty(
            seller,
            listingRecipients[listingId],
            price
        );

        emit PurchaseCompleted(
            listingId,
            tokenId,
            buyer,
            price
        );
    }
```

16. Complete project file atlas

The atlas covers 293 repository-visible source, configuration, test, documentation and evidence files captured for this snapshot. Each entry identifies its role and available code symbols with source line numbers. Test cases are intended coverage, not newly executed results. Import lists describe direct static references, not a complete runtime call graph.

A directory literally named "contracts " (with a trailing space) also exists in the listing. Its starter files are documented separately and are not assumed to be the active contracts project. Installed dependencies, generated build output and private runtime files are excluded from per-file business-logic explanations; their handling is specified in the final chapter.

Directory: Root documentation and evidence

.gitattributes

Repository/tooling metadata controlling ignores, linting or project behavior; not user-facing application logic.

Size: 66 bytes | 2 text lines

.gitignore

Repository/tooling metadata controlling ignores, linting or project behavior; not user-facing application logic.

Size: 717 bytes | 45 text lines

acceptance_results.json

Saved consolidated command outcomes from a past acceptance run; not a fresh test execution for this handbook.

Size: 669 bytes | 31 text lines

Top-level fields: started_utc, scope, checks

AGENTS.md

Developer/agent workflow instructions and tooling guidance; not executed as an application feature.

Size: 826 bytes | 14 text lines

Document sections: DKM working rules

Directory: backend

backend/acceptance.py

Runs consolidated acceptance commands and records outcomes; execution can require databases/services and should use isolated test state.

Size: 2,062 bytes | 33 text lines

Direct imports: json, subprocess, sys, time, datetime, pathlib

backend/check_demo_readiness.py

Checks configured dependencies and deployment readiness, not a substitute for interactive end-to-end purchase/download.

Size: 2,968 bytes | 59 text lines

Code entry points: main (L13)

Direct imports: json, os, pathlib, dotenv, web3, services.payments.settlement

backend/check_restore.py

Database backup/restore verification helper; inspect disposable target controls before running.

Size: 1,819 bytes | 37 text lines

Code entry points: main (L13)

Direct imports: asyncio, hashlib, json, pathlib, urllib.parse, dotenv, asyncpg, services

backend/init_content_key.py

Generates/provisions backend encryption master-key material. Output is secret and must never be pasted into this handbook.

Size: 808 bytes | 21 text lines

Code entry points: main (L8)

Direct imports: base64, os, pathlib, dotenv

backend/main.py

FastAPI composition root: registers routers, starts database setup/workers, validates uploads, performs Pinata storage, search, access/download, recovery and provider-order endpoints. The central bridge between HTTP requests and services.

Size: 32,815 bytes | 794 text lines

Code entry points: duplicate_upload (L64); validation_error (L69); private_response_headers (L77); startup (L95); shutdown (L123); root (L135); health (L140); analyze (L196); upload_ipfs (L219); SearchRequest (L401); read_upload (L408); search (L418); check_asset_access (L429); download_asset (L453); asset_description (L493); asset_details (L508); LinkTokenRequest (L545); recover_mint (L550); link_token (L577); dashboard (L602); capabilities (L614); account_reviews (L621)

Code entry points: latest_payment (L638); activity (L649); require_order_owner (L667); razorpay_webhook (L682); settle_payment (L721); create_payment (L731); get_payment (L754); confirm_payment (L773)

Direct imports: fastapi, pydantic, fastapi.exceptions, fastapi.responses, fastapi.middleware.cors, os, json, httpx, asyncio, dotenv, services.text_extractor, services.asset_price, services.llm_enrichment, services.embedding, services.pgvector_client, services.quantum_hash, services.quantum_search, services.fraud_detector, services.blockchain, services.payments.models, services.payments.service, services.auth, services.reviews, services.content_security, services.payments.transak, services.payments.quotes, services.payments.transak_discovery, services.payments.asset_funding, services.payments, services.research_results, services.payments.payouts, services.payments.recording, services.event_indexer, services.positions

HTTP surface: GET /; GET /health; POST /api/analyze; POST /api/upload-ipfs; POST /api/search; GET /api/assets/{token_id}/access; GET /api/assets/{token_id}/download; GET /api/assets/{token_id}/details; POST /api/assets/{vector_id}/recover-mint; POST /api/assets/{vector_id}/link-token; GET /api/account/dashboard; GET /api/capabilities; GET /api/account/reviews; GET /api/assets/{token_id}/payment; GET /api/account/activity; POST /api/payments/webhook/razorpay; POST /api/payments/{order_id}/settle; POST /api/payments/create; GET /api/payments/{order_id}; POST /api/payments/{order_id}/confirm

backend/migration_inventory.py

Snapshots old deployment/state for safe transition planning.

Size: 5,501 bytes | 83 text lines

Code entry points: main (L16)

Direct imports: asyncio, json, os, re, datetime, pathlib, dotenv, web3, services, services.event_indexer, services.pgvector_client

backend/operator_roles.sql

Database role/privilege setup for operational separation.

Size: 1,317 bytes | 20 text lines

backend/payment_recovery.py

Operator CLI for payment reconciliation; may change financial workflow state.

Size: 1,541 bytes | 39 text lines

Code entry points: main (L11)

Direct imports: argparse, asyncio, dotenv, services, services.payments.recovery

backend/payout_recovery.py

Operator CLI for seller payout reconciliation.

Size: 1,597 bytes | 36 text lines

Code entry points: main (L10)

Direct imports: argparse, asyncio, json, dotenv, services.payments.payouts

backend/requirements-research.txt

Python dependency constraints for research simulation. Installed versions and saved benchmark versions may differ.

Size: 148 bytes | 3 text lines

backend/requirements.txt

Python dependency constraints for backend runtime. Installed versions and saved benchmark versions may differ.

Size: 386 bytes | 21 text lines

backend/research/__init__.py

Python package initializer; may be intentionally empty.

Size: 82 bytes | 1 text lines

backend/research/qaoa_pricing.py

Synthetic demand objective, diagonal QAOA cost/mixer circuit, sampled selection and classical/exact comparators.

Size: 3,952 bytes | 70 text lines

Code entry points: demand_revenue (L8); circuit_for (L19); optimize_prices (L38)

Direct imports: numpy, pennylane, scipy.optimize, scipy.special

backend/research/run_embeddings.py

Constructs the authored corpus/query splits, runs comparable embedding methods and saves per-query metrics/artifacts.

Size: 7,433 bytes | 78 text lines

Code entry points: corpus (L23); main (L37)

Direct imports: json, time, hashlib, pathlib, importlib.metadata, numpy, sentence_transformers, vqc_embeddings

backend/research/run_pricing.py

Executes the seeded synthetic pricing experiment and writes per-case data and report.

Size: 5,441 bytes | 67 text lines

Code entry points: main (L11)

Direct imports: json, time, pathlib, importlib.metadata, numpy, scipy.optimize, qaoa_pricing

backend/research/vqc_embeddings.py

PCA/angle preprocessing, variational quantum feature circuit, training and ridge reconstruction for the separate VQC benchmark.

Size: 4,365 bytes | 89 text lines

Code entry points: normalize (L8); quantum_features (L16); decoder (L30); expand (L36); train (L40); transform (L70); ndcg (L81)

Direct imports: numpy, pennylane, sklearn.decomposition

backend/services/__init__.py

Python package initializer; may be intentionally empty.

Size: 0 bytes | binary/unreadable as UTF-8

backend/services/activity.py

Builds sanitized account activity from available persisted/indexed records. Incomplete indexing cannot establish complete lifetime history.

Size: 2,555 bytes | 49 text lines

Code entry points: account_activity (L15)

Direct imports: web3, os, ., payments.settlement

backend/services/asset_price.py

Validates exact decimal asset prices and rejects invalid precision/range instead of relying on binary floating-point money calculations.

Size: 408 bytes | 10 text lines

Code entry points: asset_price (L4)

Direct imports: decimal

backend/services/auth.py

Password accounts with scrypt, hashed revocable cookie sessions, exact-origin checks, rate limiting, Google OAuth plumbing and signed wallet challenges. Account identity and wallet identity are separate until proof is verified.

Size: 15,339 bytes | 320 text lines

Code entry points: setup_auth (L29); digest (L59); password_hash (L63); hash_password (L69); same_origin (L75); current_user (L81); wallet_user (L98); Credentials (L104); Credentials.normalize (L110); throttle (L117); new_session (L133); signup (L142); login (L159); google_config (L175); methods (L183); google_start (L188); google_callback (L210); me (L261); logout (L267); WalletProof (L278); wallet_challenge (L284); verify_wallet (L295)

Direct imports: asyncio, base64, hashlib, hmac, os, re, secrets, urllib.parse, asyncpg, httpx, eth_account, eth_account.messages, fastapi, fastapi.responses, pydantic, web3, pgvector_client

HTTP surface: POST /api/auth/signup; POST /api/auth/login; GET /api/auth/methods; POST /api/auth/google/start; GET /api/auth/google/callback; GET /api/auth/me; POST /api/auth/logout; POST /api/auth/wallet/challenge; POST /api/auth/wallet/verify

backend/services/blockchain.py

RPC contract reads and current/legacy registry and entitlement lookup. Keeps legacy access bounded to the preserved token prefix.

Size: 4,781 bytes | 159 text lines

Code entry points: check_blockchain_connection (L28); entitlement_markets (L67); has_purchased (L92); get_purchase_count (L97); get_knowledge_asset (L119); legacy_registry_address (L144); market_token_supported (L156)

Direct imports: os, web3, web3.middleware, dotenv, knowledge_asset_abi, marketplace_abi

backend/services/chain_event_abis.json

Contract ABI/interface data used to encode calls and decode outputs/events. An ABI is not a deployment or evidence of a successful transaction.

Size: 36,396 bytes | 1,667 text lines

Top-level fields: marketplace, splitter

backend/services/content_security.py

AES-256-GCM envelope encryption and decryption. Uses independent random nonces for document encryption and key wrapping, with the plaintext digest as associated authenticated data.

Size: 1,192 bytes | 31 text lines

Code entry points: master_key (L7); seal (L17); unseal (L27)

Direct imports: base64, os, cryptography.hazmat.primitives.ciphers.aead

backend/services/embedding.py

Lazy local-cache loading of all-MiniLM-L6-v2; produces normalized 384-dimensional summary/query vectors. It does not call Ollama to produce embeddings.

Size: 729 bytes | 30 text lines

Code entry points: get_model (L6); generate_embedding (L18); embed_query (L29)

Direct imports: sentence_transformers

backend/services/event_indexer.py

Persists chain-event sources, checkpoints and derived activity. Handles idempotency and chain consistency; source availability and backlog determine coverage.

Size: 14,897 bytes | 234 text lines

Code entry points: setup_indexer (L22); first_code_block (L49); normalized (L59); sale_accounting (L67); fetch_page (L92); sync_market (L139); rebuild_market (L186); index_worker (L197); history (L215)

Direct imports: asyncio, json, logging, os, datetime, pathlib, fastapi, web3, web3._utils.events, eth_utils, services.auth, services.pgvector_client, services

HTTP surface: GET /api/account/indexed-activity

backend/services/fraud_detector.py

Checks persisted BLAKE3 duplicates and files smaller than 100 bytes. Despite its class name, this is a classical rule-based screen, not learned fraud or plagiarism detection.

Size: 1,280 bytes | 37 text lines

Code entry points: QuantumFraudDetector (L5); QuantumFraudDetector.check_upload (L6)

Direct imports: services.quantum_hash, services

backend/services/knowledge_asset_abi.py

Contract ABI/interface data used to encode calls and decode outputs/events. An ABI is not a deployment or evidence of a successful transaction.

Size: 2,505 bytes | 82 text lines

backend/services/llm_enrichment.py

Calls Ollama through the OpenAI-compatible client, validates structured metadata and four evidence-backed criteria, then computes quality on the server. Rejects fabricated or malformed support rather than manufacturing a fallback score.

Size: 6,486 bytes | 116 text lines

Code entry points: AnalysisUnavailable (L52); validate_metadata (L56); enrich_document (L97)

Direct imports: asyncio, json, os, openai

backend/services/marketplace_abi.py

Contract ABI/interface data used to encode calls and decode outputs/events. An ABI is not a deployment or evidence of a successful transaction.

Size: 1,222 bytes | 52 text lines

backend/services/mint_recovery.py

Finds already minted registrations for a saved upload and prevents an immediate blind retry when completion can be recovered. Uses creator/content matching and bounded scanning.

Size: 735 bytes | 11 text lines

Code entry points: find_mint (L2)

backend/services/payments/asset_funding.py

Binds funding intent to asset/market/token context; funding alone must not grant access.

Size: 8,830 bytes | 162 text lines

Code entry points: setup_funding (L19); nominal (L36); public (L42); FundingRequest (L48); funding_quote (L54); funding_session (L71); reconcile_funding (L96); create (L139); session (L148); history (L157)

Direct imports: asyncio, json, time, decimal, uuid, fastapi, pydantic, services, services.auth, services.pgvector_client, ., quotes, market_profiles, token_delivery

backend/services/payments/capabilities.py

Reads deployed market/token wiring and reports route availability. Fails closed when chain/profile verification is unavailable.

Size: 5,133 bytes | 84 text lines

Code entry points: chain_routes (L13); payment_capabilities (L40)

Direct imports: os, web3, ., razorpay_provider, settlement, market_profiles, web3.middleware

backend/services/payments/market_profiles.py

Reads token symbol/decimals and market wiring from deployed contracts rather than assuming every token called USDC is identical.

Size: 3,733 bytes | 65 text lines

Code entry points: parse_units (L12); market_profile (L27)

Direct imports: re, web3, settlement

backend/services/payments/mock_provider.py

Non-spending payment simulator for tests. Its confirmation must not become blockchain entitlement. Distinct from the real on-testnet MockERC20 contract.

Size: 629 bytes | 28 text lines

Code entry points: MockPaymentProvider (L7); MockPaymentProvider.create_payment (L9); MockPaymentProvider.verify_payment (L19)

Direct imports: uuid, models, provider

backend/services/payments/models.py

Typed payment methods, statuses and request/order models. Defines the vocabulary used by persistence and provider state transitions.

Size: 2,224 bytes | 74 text lines

Code entry points: PaymentStatus (L8); PaymentMethod (L18); CreatePaymentRequest (L25); CreatePaymentRequest.validate_buyer (L32); PaymentOrder (L38); ConfirmPaymentRequest (L71)

Direct imports: enum, typing, pydantic, web3

backend/services/payments/payouts.py

Retained seller payout preferences and order tracking; preferences do not themselves perform conversion or bank payment.

Size: 11,087 bytes | 193 text lines

Code entry points: setup_payouts (L14); digest (L48); create_payout (L52); verified_sell (L68); refresh_payout (L89); resolve_payout (L131); reconcile_payouts (L152); PayoutLookup (L170); refresh (L176); history (L185)

Direct imports: hashlib, json, uuid, httpx, fastapi, pydantic, services.auth, services.payments, services.pgvector_client

backend/services/payments/provider.py

Provider interface separating external payment operations from order orchestration.

Size: 609 bytes | 27 text lines

Code entry points: PaymentProvider (L6); PaymentProvider.create_payment (L9); PaymentProvider.verify_payment (L19)

Direct imports: abc, models

backend/services/payments/quotes.py

Builds purchase terms and validates buyer, market, token, amount and expiry. A quote is not an asset reservation or a payment receipt.

Size: 7,803 bytes | 142 text lines

Code entry points: setup_quotes (L22); accounting (L36); purchase_terms (L51); public_quote (L88); create_quote (L94); accept_quote (L105); create (L126); accept (L135)

Direct imports: asyncio, json, datetime, uuid, fastapi, web3, services.auth, services.pgvector_client, services, settlement, market_profiles

HTTP surface: POST /api/purchase-quotes/asset/{token_id}; POST /api/purchase-quotes/{quote_id}/accept

backend/services/payments/razorpay_provider.py

Retained Razorpay test-mode creation/signature/status integration. Razorpay collection itself is not INR-to-USDC conversion.

Size: 3,347 bytes | 61 text lines

Code entry points: RazorpayProvider (L13); RazorpayProvider.__init__ (L14); RazorpayProvider.quote (L18); RazorpayProvider.create_payment (L32); RazorpayProvider.verify_payment (L43)

Direct imports: hashlib, hmac, os, decimal, httpx, models, provider

backend/services/payments/receipt_registry_abi.json

Contract ABI/interface data used to encode calls and decode outputs/events. An ABI is not a deployment or evidence of a successful transaction.

Size: 10,757 bytes | 498 text lines

Structured array with 19 entries.

backend/services/payments/recording.py

Optional blockchain receipt recorder workflow with durable transaction management. Receipt logging is not a source of NFT access entitlement.

Size: 16,555 bytes | 284 text lines

Code entry points: setup_recording (L28); canonical (L55); commitment (L59); payment_tuple (L63); payment_hash (L68); enqueue_verified (L72); enqueue_reversal (L114); connection (L136); call_for (L152); prepare (L159); validate_source (L168); verify_or_broadcast (L182); record_job (L227); recording_worker (L260); history (L274)

Direct imports: asyncio, hashlib, hmac, json, logging, os, re, pathlib, uuid, eth_abi, fastapi, web3, web3.middleware, web3.exceptions, services.auth, services.content_security, services.pgvector_client, settlement

HTTP surface: GET /api/payments/recordings

backend/services/payments/recording_sources.py

Validates external/provider/chain source evidence before an optional recording job may assert a receipt.

Size: 6,438 bytes | 90 text lines

Code entry points: queue_indexed_sales (L13); verify_saved_capture (L56); saved_payment (L62); queue_verified_refunds (L76)

Direct imports: asyncio, json, os, re, httpx, web3, services, services.pgvector_client, recording

backend/services/payments/recovery.py

Operator reconciliation helpers for unsettled/captured orders, retries and refund-related state; execution can have external effects.

Size: 5,590 bytes | 90 text lines

Code entry points: require_refundable (L14); verify_refund (L31); recover (L57)

Direct imports: asyncio, re, httpx, web3, services, ., models, razorpay_provider

backend/services/payments/sale_payouts.py

Binds payout eligibility to verified canonical sales, correct seller, token and positive proceeds.

Size: 6,831 bytes | 106 text lines

Code entry points: SaleQuote (L16); verified_sale (L24); create_quote (L44); open_session (L63); outgoing_delivery (L86); create (L96); session (L103)

Direct imports: asyncio, json, time, uuid, fastapi, pydantic, services, services.auth, services.pgvector_client, ., asset_funding, token_delivery

backend/services/payments/service.py

Persists and transitions payment orders and coordinates providers; retains idempotency and confirmation/settlement separation.

Size: 8,193 bytes | 154 text lines

Code entry points: PaymentService (L14); PaymentService.__init__ (L15); PaymentService.create_order (L21); PaymentService.get_order (L91); PaymentService.confirm_order (L99); PaymentService._save (L145)

Direct imports: uuid, asyncio, os, httpx, decimal, services.pgvector_client, mock_provider, models, razorpay_provider

backend/services/payments/settlement.py

Verifies configured chain/contract and performs controlled funded on-chain settlement, with receipt/state verification. A prefunded treasury is not currency exchange.

Size: 15,361 bytes | 282 text lines

Code entry points: function_abi (L17); order_amount (L38); verify_denomination (L46); connection (L55); validate_checkout (L68); prepare (L100); confirmed_receipt (L115); broadcast_and_verify (L128); has_entitlement (L160); settle_order (L166);

reconcile_payments (L255); settlement_worker (L275)

Direct imports: asyncio, os, logging, datetime, web3, web3.middleware, web3.exceptions, services.pgvector_client, services.knowledge_asset_abi, services.marketplace_abi, models

backend/services/payments/token_delivery.py

Checks actual canonical token delivery evidence separately from provider completion status.

Size: 2,688 bytes | 46 text lines

Code entry points: verify_delivery (L9)

Direct imports: re, web3

backend/services/payments/transak.py

Retained Transak staging order/session verification and durable workflow code. Not the current wallet-only UI payment route.

Size: 17,863 bytes | 343 text lines

Code entry points: setup_transak (L25); configuration (L48); provider_json (L57); access_token (L70); create_session (L86); status (L130); session (L141); OrderLookup (L151); verified_order (L155); normalized_order (L165); reconcile_receipt (L194); refresh_order (L212); check_order (L248); orders (L257); reconcile_saved_orders (L268); reconciliation_worker (L296); PayoutPreference (L307); payout_preference (L313); save_payout_preference (L324); payout_session (L342)

Direct imports: hashlib, asyncio, logging, ipaddress, json, re, decimal, uuid, pydantic, os, time, urllib.parse, httpx, fastapi, services.auth, services.content_security, services.pgvector_client

HTTP surface: GET /api/payments/transak/status; POST /api/payments/transak/session; POST /api/payments/transak/orders/refresh; GET /api/payments/transak/orders; GET /api/payments/transak/payout-preference; PUT /api/payments/transak/payout-preference; POST /api/payments/transak/payout-session

backend/services/payments/transak_discovery.py

Retained provider discovery and quote coverage lookup. Returned methods depend on the provider; no universal currency support is implied.

Size: 6,733 bytes | 128 text lines

Code entry points: text (L17); amount (L23); lookup (L33); discover (L42); crypto_coverage (L71); quote (L86); coverage (L123)

Direct imports: asyncio, ipaddress, re, time, decimal, httpx, fastapi, services.auth, transak

HTTP surface: GET /api/payments/transak/coverage

backend/services/pgvector_client.py

PostgreSQL schema setup, compatible column migrations, asset persistence, duplicate protection, vector decoding and cosine candidate queries. Owns the knowledge_assets catalog and initial payment/review tables.

Size: 12,850 bytes | 385 text lines

Code entry points: DuplicateUploadError (L9); get_conn (L18); setup_tables (L23); insert_embedding (L174); semantic_search (L238); update_token_id (L325); get_asset_by_token_id (L337)

Direct imports: os, asyncpg, numpy, pgvector.asyncpg, uuid, decimal

backend/services/positions.py

Reads wallet-specific on-chain positions including owned assets, listings, auctions and refundable bids, preserving market/registry identity.

Size: 3,322 bytes | 50 text lines

Code entry points: snapshot (L12); positions (L48)

Direct imports: asyncio, fastapi, web3, ., auth, event_indexer

HTTP surface: GET /api/account/positions

backend/services/quantum_hash.py

Classical BLAKE3 digest and integrity comparison. The historical quantum_hash name does not make the implementation a quantum algorithm.

Size: 903 bytes | 36 text lines

Code entry points: quantum_hash (L4); verify_integrity (L11); hash_string (L18)

Direct imports: blake3

backend/services/quantum_search.py

Current cosine-candidate plus deterministic MMR search, with legacy graph propagation and annealing helpers retained separately. Production search does not call QAOA or VQC.

Size: 5,270 bytes | 123 text lines

Code entry points: build_asset_graph (L8); quantum_walk_discover (L26); qubo_rerank (L56); quantum_enhanced_search (L86); rank_results (L97)

Direct imports: asyncio, networkx, numpy, dimod

backend/services/research_results.py

Authenticated API exposing retained experiment reports/results without executing quantum experiments on every request.

Size: 1,593 bytes | 33 text lines

Code entry points: load_result (L11); results (L22)

Direct imports: json, pathlib, fastapi, services.auth

HTTP surface: GET /api/research/results

backend/services/reviews.py

Review APIs bound to verified wallet/account and purchase entitlement; includes legacy entitlement compatibility.

Size: 3,047 bytes | 56 text lines

Code entry points: Review (L11); get_reviews (L17); save_review (L34)

Direct imports: asyncio, fastapi, pydantic, ., auth

HTTP surface: GET /api/assets/{token_id}/reviews; POST /api/assets/{token_id}/reviews

backend/services/text_extractor.py

Converts supported uploaded formats into bounded text excerpts for analysis. Extraction is partial, not a complete document-understanding guarantee.

Size: 1,599 bytes | 63 text lines

Code entry points: extract_text (L5); _extract_from_pdf (L22); _extract_from_csv (L46); _extract_from_text (L59)

Direct imports: io, pdfplumber

backend/sync_history.py

Backfills event history from chain with bounded/checkpointed progress.

Size: 1,289 bytes | 31 text lines

Code entry points: main (L9)

Direct imports: argparse, asyncio, dotenv, services

backend/test_blockchain.py

Test or diagnostic coverage for blockchain. The extracted cases below describe intended checks; file presence alone does not prove that they passed.

Size: 237 bytes | 8 text lines

Direct imports: services.blockchain

backend/test_insert.py

Test or diagnostic coverage for insert. The extracted cases below describe intended checks; file presence alone does not prove that they passed.

Size: 602 bytes | 26 text lines

Code entry points: main (L8)

Direct imports: asyncio, services.embedding, services.pgvector_client, services.quantum_hash

backend/test_knowledge_asset.py

Test or diagnostic coverage for knowledge asset. The extracted cases below describe intended checks; file presence alone does not prove that they passed.

Size: 657 bytes | 21 text lines

Direct imports: services.blockchain

backend/test_marketplace.py

Test or diagnostic coverage for marketplace. The extracted cases below describe intended checks; file presence alone does not prove that they passed.

Size: 481 bytes | 24 text lines

Direct imports: services.blockchain

backend/test_raw_vector.py

Test or diagnostic coverage for raw vector. The extracted cases below describe intended checks; file presence alone does not prove that they passed.

Size: 556 bytes | 29 text lines

Code entry points: main (L7)

Direct imports: asyncio, services.embedding, services.pgvector_client

backend/test_search.py

Test or diagnostic coverage for search. The extracted cases below describe intended checks; file presence alone does not prove that they passed.

Size: 1,753 bytes | 60 text lines

Code entry points: main (L9)

Direct imports: argparse, asyncio, sys, services.embedding, services.pgvector_client

backend/test_setup.py

Test or diagnostic coverage for setup. The extracted cases below describe intended checks; file presence alone does not prove that they passed.

Size: 102 bytes | 6 text lines

Direct imports: asyncio, services.pgvector_client

backend/tests/__init__.py

Test or diagnostic coverage for init . The extracted cases below describe intended checks; file presence alone does not prove that they passed.

Size: 0 bytes | binary/unreadable as UTF-8

backend/tests/test_activity.py

Test or diagnostic coverage for activity. The extracted cases below describe intended checks; file presence alone does not prove that they passed.

Size: 2,688 bytes | 44 text lines

Code entry points: ActivityTests (L6); ActivityTests.test_bounded_chain_history_filters_by_wallet (L7); ActivityTests.test_history_includes_reviewed_previous_markets (L28)

Direct imports: unittest, unittest.mock, services

Declared checks: ActivityTests.test_bounded_chain_history_filters_by_wallet; ActivityTests.test_history_includes_reviewed_previous_markets

backend/tests/test_asset_funding.py

Test or diagnostic coverage for asset funding. The extracted cases below describe intended checks; file presence alone does not prove that they passed.

Size: 4,769 bytes | 69 text lines

Code entry points: FundingTests (L12); FundingTests.asyncSetUp (L14);
FundingTests.test_binding_freshness_and_concurrent_session_attempt (L27);
FundingTests.test_completed_provider_is_not_chain_delivery_or_entitlement (L42)

Direct imports: asyncio, json, time, unittest, asyncpg, uuid, unittest.mock, tests, services.payments

Declared checks: FundingTests.test_binding_freshness_and_concurrent_session_attempt;
FundingTests.test_completed_provider_is_not_chain_delivery_or_entitlement

backend/tests/test_asset_price.py

Test or diagnostic coverage for asset price. The extracted cases below describe intended checks; file presence alone does not prove that they passed.

Size: 671 bytes | 15 text lines

Code entry points: AssetPriceTests (L6); AssetPriceTests.test_cents_and_existing_whole_prices_are_exact (L7);
AssetPriceTests.test_invalid_prices_are_rejected (L12)

Direct imports: unittest, decimal, services.asset_price

Declared checks: AssetPriceTests.test_cents_and_existing_whole_prices_are_exact; AssetPriceTests.test_invalid_prices_are_rejected

backend/tests/test_auth.py

Test or diagnostic coverage for auth. The extracted cases below describe intended checks; file presence alone does not prove that they passed.

Size: 38,085 bytes | 557 text lines

Code entry points: AuthTests (L17); AuthTests.test_recover_mint_is_account_scoped_and_revalidates_chain_content (L18);
AuthTests.test_price_migration_preserves_existing_uploads_and_cents_search (L50);
AuthTests.test_concurrent_auth_abuse_limit_is_atomic (L66); AuthTests.asyncSetUp (L81); AuthTests.asyncTearDown (L106);
AuthTests.signup (L112); AuthTests.verify (L117); AuthTests.test_sessions_csrf_passwords_and_logout (L125);
AuthTests.test_wallet_proof_replay_and_account_isolation (L156);
AuthTests.test_link_checks_content_and_access_uses_session_wallet (L193); AuthTests.test_expired_wrong_proof_and_rate_limit
(L211); AuthTests.test_upload_form_and_returned_id_belong_to_account (L224);
AuthTests.test_google_code_flow_creates_and_reuses_separate_account (L254);
AuthTests.test_asset_details_only_returns_catalog_fields (L299);
AuthTests.test_persistent_duplicate_detection_concurrency_and_retry (L312);
AuthTests.test_protected_download_checks_entitlement_and_ciphertext (L328);
AuthTests.test_reviews_require_purchase_and_update_one_rating (L375);
AuthTests.test_search_filters_and_activity_account_boundary (L410);
AuthTests.test_google_rejects_missing_config_state_cookie_expiry_and_unverified_identity (L435);
AuthTests.test_transak_requires_session_origin_verified_wallet_and_never_grants_access (L462);
AuthTests.test_transak_history_requires_account_and_verified_refresh (L480)

Code entry points: AuthTests.test_seller_preference_is_scoped_and_not_payout_authorization (L502);
AuthTests.test_purchase_quotes_require_session_wallet_and_origin (L538)

Direct imports: uuid, unittest, unittest.mock, httpx, dotenv, eth_account, eth_account.messages, services, services.payments, main

Declared checks: AuthTests.test_recover_mint_is_account_scoped_and_revalidates_chain_content;
AuthTests.test_price_migration_preserves_existing_uploads_and_cents_search;
AuthTests.test_concurrent_auth_abuse_limit_is_atomic; AuthTests.test_sessions_csrf_passwords_and_logout;
AuthTests.test_wallet_proof_replay_and_account_isolation; AuthTests.test_link_checks_content_and_access_uses_session_wallet;
AuthTests.test_expired_wrong_proof_and_rate_limit; AuthTests.test_upload_form_and_returned_id_belong_to_account;
AuthTests.test_google_code_flow_creates_and_reuses_separate_account; AuthTests.test_asset_details_only_returns_catalog_fields;
AuthTests.test_persistent_duplicate_detection_concurrency_and_retry;
AuthTests.test_protected_download_checks_entitlement_and_ciphertext

Declared checks: AuthTests.test_reviews_require_purchase_and_update_one_rating;
AuthTests.test_search_filters_and_activity_account_boundary;
AuthTests.test_google_rejects_missing_config_state_cookie_expiry_and_unverified_identity;
AuthTests.test_transak_requires_session_origin_verified_wallet_and_never_grants_access;
AuthTests.test_transak_history_requires_account_and_verified_refresh;

AuthTests.test_seller_preference_is_scoped_and_not_payout_authorization;
AuthTests.test_purchase_quotes_require_session_wallet_and_origin

backend/tests/test_capabilities.py

Test or diagnostic coverage for capabilities. The extracted cases below describe intended checks; file presence alone does not prove that they passed.

Size: 2,247 bytes | 32 text lines

Code entry points: CapabilityTests (L7); CapabilityTests.test_unknown_chain_and_missing_configuration_fail_closed (L8);
CapabilityTests.test_funding_is_not_purchase_and_old_deployment_cannot_enable_fiat (L15);
CapabilityTests.test_paused_market_does_not_disable_auction_recovery (L27)

Direct imports: unittest, unittest.mock, services.payments

Declared checks: CapabilityTests.test_unknown_chain_and_missing_configuration_fail_closed;
CapabilityTests.test_funding_is_not_purchase_and_old_deployment_cannot_enable_fiat;
CapabilityTests.test_paused_market_does_not_disable_auction_recovery

backend/tests/test_indexer.py

Test or diagnostic coverage for indexer. The extracted cases below describe intended checks; file presence alone does not prove that they passed.

Size: 6,547 bytes | 101 text lines

Code entry points: IndexTests (L15); IndexTests.test_idempotent_history_accounting_and_reorg_fail_closed (L19);
IndexTests.test_deployment_binary_search_and_large_integers (L48);
IndexTests.test_rpc_failure_shrinks_page_and_persists_backoff_without_advancing (L53);
IndexTests.test_auction_and_refund_logs_decode_with_market_identity (L71)

Direct imports: asyncio, json, unittest, types, unittest.mock, tests, services, web3, web3.datastructures, eth_abi, eth_utils, hexbytes

Declared checks: IndexTests.test_idempotent_history_accounting_and_reorg_fail_closed;
IndexTests.test_deployment_binary_search_and_large_integers;
IndexTests.test_rpc_failure_shrinks_page_and_persists_backoff_without_advancing;
IndexTests.test_auction_and_refund_logs_decode_with_market_identity

backend/tests/test_legacy_access.py

Test or diagnostic coverage for legacy access. The extracted cases below describe intended checks; file presence alone does not prove that they passed.

Size: 3,451 bytes | 57 text lines

Code entry points: LegacyAccessTests (L10); LegacyAccessTests.test_successor_filters_legacy_entitlements_above_snapshot (L11);
LegacyAccessTests.test_old_purchase_counts_and_registry_validation (L33)

Direct imports: os, unittest, unittest.mock, dotenv, services

Declared checks: LegacyAccessTests.test_successor_filters_legacy_entitlements_above_snapshot;
LegacyAccessTests.test_old_purchase_counts_and_registry_validation

backend/tests/test_llm_enrichment.py

Test or diagnostic coverage for llm enrichment. The extracted cases below describe intended checks; file presence alone does not prove that they passed.

Size: 2,965 bytes | 50 text lines

Code entry points: metadata (L12); EnrichmentTests (L19); EnrichmentTests.test_total_computed_from_rubric_not_model_total (L20);
EnrichmentTests.test_reject_fabricated_evidence_invalid_scores_and_missing_criteria (L29);
EnrichmentTests.test_ollama_options_and_no_fabricated_fallback (L38); EnrichmentTests.test_bad_pdf_returns_no_content (L49)

Direct imports: json, unittest, types, unittest.mock, services, services.text_extractor

Declared checks: EnrichmentTests.test_total_computed_from_rubric_not_model_total;
EnrichmentTests.test_reject_fabricated_evidence_invalid_scores_and_missing_criteria;
EnrichmentTests.test_ollama_options_and_no_fabricated_fallback; EnrichmentTests.test_bad_pdf_returns_no_content

backend/tests/test_market_profiles.py

Test or diagnostic coverage for market profiles. The extracted cases below describe intended checks; file presence alone does not prove that they passed.

Size: 3,356 bytes | 55 text lines

Code entry points: MarketProfileTests (L9); MarketProfileTests.test_exact_amounts_and_invalid_inputs (L10); MarketProfileTests.test_profiles_preserve_distinct_denominations_and_fail_closed (L19); MarketProfileTests.test_verification_rejects_wrong_chain_and_missing_code (L43); MarketProfileTests.test_persisted_legacy_and_new_amounts_do_not_reinterpret_units (L51)

Direct imports: unittest, types, unittest.mock, services.payments.market_profiles, verify_transak_token

Declared checks: MarketProfileTests.test_exact_amounts_and_invalid_inputs; MarketProfileTests.test_profiles_preserve_distinct_denominations_and_fail_closed; MarketProfileTests.test_verification_rejects_wrong_chain_and_missing_code; MarketProfileTests.test_persisted_legacy_and_new_amounts_do_not_reinterpret_units

backend/tests/test_mint_recovery.py

Test or diagnostic coverage for mint recovery. The extracted cases below describe intended checks; file presence alone does not prove that they passed.

Size: 1,469 bytes | 26 text lines

Code entry points: MintRecoveryTests (L5); MintRecoveryTests.test_matches_both_content_ids_and_creator_at_same_block (L6); MintRecoveryTests.test_rpc_failure_and_oversized_catalog_never_report_not_minted (L21)

Direct imports: unittest, unittest.mock, services.mint_recovery

Declared checks: MintRecoveryTests.test_matches_both_content_ids_and_creator_at_same_block; MintRecoveryTests.test_rpc_failure_and_oversized_catalog_never_report_not_minted

backend/tests/test_payments.py

Test or diagnostic coverage for payments. The extracted cases below describe intended checks; file presence alone does not prove that they passed.

Size: 24,802 bytes | 415 text lines

Code entry points: PaymentTests (L25); PaymentTests.asyncSetUp (L26); PaymentTests.asyncTearDown (L66); PaymentTests.test_restart_and_concurrent_confirmation (L73); PaymentTests.test_concurrent_checkout_creation_reuses_one_provider_order (L99); PaymentTests.test_settlement_verifies_hash_receipt_depth_sender_and_call (L112); PaymentTests.test_invalid_and_failed_orders (L147); PaymentTests.test_settlement_checkpoint_and_retry (L165); PaymentTests.test_chain_success_without_entitlement_stays_settled (L198); PaymentTests.test_razorpay_signature_capture_and_quote (L212); PaymentTests.test_recovery_budget_and_refund_are_durable (L237); PaymentTests.test_recovery_timeout_keeps_signed_transaction (L269); PaymentTests.test_refund_rejects_pending_successful_and_shallow_receipts (L283); PaymentTests.test_refund_checks_provider_amount_identity_and_status (L311); PaymentTests.test_operator_role_records_session_identity_and_cannot_rewrite_audit (L332); PaymentTests.test_wallet_validation (L359); PaymentTests.test_mined_revert_requires_explicit_retry (L364); PaymentTests.test_bad_callback_does_not_poison_pending_order (L387); PaymentTests.test_webhook_checks_raw_body_signature (L395)

Direct imports: asyncio, uuid, unittest, os, hashlib, hmac, httpx, unittest.mock, dotenv, services, services.payments, services.payments.razorpay_provider, services.payments.models

Declared checks: PaymentTests.test_restart_and_concurrent_confirmation; PaymentTests.test_concurrent_checkout_creation_reuses_one_provider_order; PaymentTests.test_settlement_verifies_hash_receipt_depth_sender_and_call; PaymentTests.test_invalid_and_failed_orders; PaymentTests.test_settlement_checkpoint_and_retry; PaymentTests.test_chain_success_without_entitlement_stays_settled; PaymentTests.test_razorpay_signature_capture_and_quote; PaymentTests.test_recovery_budget_and_refund_are_durable; PaymentTests.test_recovery_timeout_keeps_signed_transaction; PaymentTests.test_refund_rejects_pending_successful_and_shallow_receipts; PaymentTests.test_refund_checks_provider_amount_identity_and_status; PaymentTests.test_operator_role_records_session_identity_and_cannot_rewrite_audit

Declared checks: PaymentTests.test_wallet_validation; PaymentTests.test_mined_revert_requires_explicit_retry; PaymentTests.test_bad_callback_does_not_poison_pending_order; PaymentTests.test_webhook_checks_raw_body_signature

backend/tests/test_payouts.py

Test or diagnostic coverage for payouts. The extracted cases below describe intended checks; file presence alone does not prove that they passed.

Size: 4,852 bytes | 92 text lines

Code entry points: PayoutTests (L9); PayoutTests.prepare (L13); PayoutTests.refresh (L25); PayoutTests.test_binding_duplicates_and_safe_fields (L28); PayoutTests.test_changed_amount_resolution_and_currency_cannot_be_overridden (L45); PayoutTests.test_terminal_regression_late_completion_and_restart_outage (L66)

Direct imports: asyncio, json, unittest, unittest.mock, services.payments, tests

Declared checks: PayoutTests.test_binding_duplicates_and_safe_fields; PayoutTests.test_changed_amount_resolution_and_currency_cannot_be_overridden; PayoutTests.test_terminal_regression_late_completion_and_restart_outage

backend/tests/test_phase1_complete.py

Test or diagnostic coverage for phase1 complete. The extracted cases below describe intended checks; file presence alone does not prove that they passed.

Size: 5,623 bytes | 181 text lines

Code entry points: test_complete_phase1_pipeline (L31)

Direct imports: pytest, services.text_extractor, services.llm_enrichment, services.embedding, services.pgvector_client, services.quantum_hash, services.quantum_search

Declared checks: test_complete_phase1_pipeline

backend/tests/test_positions.py

Test or diagnostic coverage for positions. The extracted cases below describe intended checks; file presence alone does not prove that they passed.

Size: 2,558 bytes | 37 text lines

Code entry points: PositionTests (L7); PositionTests.test_current_and_legacy_positions_are_account_scoped_and_pinned (L8)

Direct imports: unittest, types, unittest.mock, services

Declared checks: PositionTests.test_current_and_legacy_positions_are_account_scoped_and_pinned

backend/tests/test_qaoa.py

Test or diagnostic coverage for qaoa. The extracted cases below describe intended checks; file presence alone does not prove that they passed.

Size: 1,435 bytes | 29 text lines

Code entry points: QAOATests (L6); QAOATests.test_zero_angles_uniform_and_one_qubit_known_optimum (L7); QAOATests.test_seeded_results_normalize_and_respect_feasible_grid (L13); QAOATests.test_invalid_inputs_and_fee_application (L22)

Direct imports: unittest, numpy, research.qaoa_pricing

Declared checks: QAOATests.test_zero_angles_uniform_and_one_qubit_known_optimum; QAOATests.test_seeded_results_normalize_and_respect_feasible_grid; QAOATests.test_invalid_inputs_and_fee_application

backend/tests/test_quotes.py

Test or diagnostic coverage for quotes. The extracted cases below describe intended checks; file presence alone does not prove that they passed.

Size: 4,481 bytes | 82 text lines

Code entry points: AccountingTests (L14); AccountingTests.test_exact_solidity_rounding_and_no_buyer_markup (L15); QuoteTests (L26); QuoteTests.asyncSetUp (L27); QuoteTests.asyncTearDown (L47); QuoteTests.test_account_isolation_and_concurrent_idempotent_accept (L52); QuoteTests.test_changes_expiry_and_failed_revalidation_do_not_accept (L67)

Direct imports: asyncio, json, unittest, uuid, unittest.mock, fastapi, dotenv, services, services.payments

Declared checks: AccountingTests.test_exact_solidity_rounding_and_no_buyer_markup; QuoteTests.test_account_isolation_and_concurrent_idempotent_accept;

QuoteTests.test_changes_expiry_and_failed_revalidation_do_not_accept

backend/tests/test_receipts.py

Test or diagnostic coverage for receipts. The extracted cases below describe intended checks; file presence alone does not prove that they passed.

Size: 1,873 bytes | 36 text lines

Code entry points: ReceiptTests (L10); ReceiptTests.test_canonical_receipt_and_reorg_boundaries (L11)

Direct imports: unittest, types, unittest.mock, web3, web3.exceptions, services.payments.settlement

Declared checks: ReceiptTests.test_canonical_receipt_and_reorg_boundaries

backend/tests/test_recording.py

Test or diagnostic coverage for recording. The extracted cases below describe intended checks; file presence alone does not prove that they passed.

Size: 10,020 bytes | 146 text lines

Code entry points: RecordingTests (L19); RecordingTests.prepare (L23); RecordingTests.test_immutable_duplicate_evidence_and_reversal (L35); RecordingTests.test_source_must_remain_confirmed_and_canonical (L54); RecordingTests.test_source_reorg_stops_before_signing (L68); RecordingTests.test_checkpoint_restart_nonce_serialization_and_reorg (L78); RecordingTests.test_payment_hash_commits_to_all_fields (L106); RecordingTests.test_signed_checkpoint_and_emitted_event_verified_before_recorded (L112)

Direct imports: asyncio, json, unittest, unittest.mock, uuid, types, web3, web3.datastructures, web3.exceptions, eth_account, eth_abi, eth_utils, hexbytes, services.payments, tests

Declared checks: RecordingTests.test_immutable_duplicate_evidence_and_reversal; RecordingTests.test_source_must_remain_confirmed_and_canonical; RecordingTests.test_source_reorg_stops_before_signing; RecordingTests.test_checkpoint_restart_nonce_serialization_and_reorg; RecordingTests.test_payment_hash_commits_to_all_fields; RecordingTests.test_signed_checkpoint_and_emitted_event_verified_before_recorded

backend/tests/test_recording_sources.py

Test or diagnostic coverage for recording sources. The extracted cases below describe intended checks; file presence alone does not prove that they passed.

Size: 4,912 bytes | 72 text lines

Code entry points: SourceTests (L10); SourceTests.prepare (L14); SourceTests.test_only_reverified_market_sale_enqueues_once_and_wrong_block_fails (L29); SourceTests.test_provider_capture_does_not_accept_authorized_refunded_or_wrong_amount (L47); SourceTests.test_failed_oldest_sale_does_not_starve_other_sales (L60)

Direct imports: json, unittest, unittest.mock, uuid, services, services.payments, tests

Declared checks: SourceTests.test_only_reverified_market_sale_enqueues_once_and_wrong_block_fails; SourceTests.test_provider_capture_does_not_accept_authorized_refunded_or_wrong_amount; SourceTests.test_failed_oldest_sale_does_not_starve_other_sales

backend/tests/test_research_results.py

Test or diagnostic coverage for research results. The extracted cases below describe intended checks; file presence alone does not prove that they passed.

Size: 2,122 bytes | 46 text lines

Code entry points: ResearchResultsTests (L12); ResearchResultsTests.test_authentication_and_generated_evidence (L13); ResearchResultsTests.test_missing_invalid_and_oversized_files (L37)

Direct imports: json, tempfile, unittest, pathlib, unittest.mock, fastapi, httpx, services

Declared checks: ResearchResultsTests.test_authentication_and_generated_evidence; ResearchResultsTests.test_missing_invalid_and_oversized_files

backend/tests/test_sale_payouts.py

Test or diagnostic coverage for sale payouts. The extracted cases below describe intended checks; file presence alone does not prove that they passed.

Size: 5,322 bytes | 72 text lines

Code entry points: SaleEvidenceTests (L12); SaleEvidenceTests.test_only_canonical_matching_seller_trnsk_proceeds_qualify (L13); SalePayoutTests (L32); SalePayoutTests.asyncSetUp (L34); SalePayoutTests.test_quote_refresh_and_only_one_sale_session (L48); SalePayoutTests.test_sell_binding_rejects_wrong_token_amount_network_and_currency (L58)

Direct imports: asyncio, json, time, unittest, uuid, unittest.mock, types, tests, services.payments

Declared checks: SaleEvidenceTests.test_only_canonical_matching_seller_trnsk_proceeds_qualify; SalePayoutTests.test_quote_refresh_and_only_one_sale_session; SalePayoutTests.test_sell_binding_rejects_wrong_token_amount_network_and_currency

backend/tests/test_search_ranking.py

Test or diagnostic coverage for search ranking. The extracted cases below describe intended checks; file presence alone does not prove that they passed.

Size: 2,089 bytes | 40 text lines

Code entry points: asset (L7); SearchRankingTests (L14); SearchRankingTests.test_duplicates_drafts_and_private_fields (L15); SearchRankingTests.test_relevance_diversity_and_repeatability (L25); SearchRankingTests.test_live_entry_point_filters_before_candidate_limit (L33)

Direct imports: unittest, unittest.mock, services.quantum_search

Declared checks: SearchRankingTests.test_duplicates_drafts_and_private_fields; SearchRankingTests.test_relevance_diversity_and_repeatability; SearchRankingTests.test_live_entry_point_filters_before_candidate_limit

backend/tests/test_token_delivery.py

Test or diagnostic coverage for token delivery. The extracted cases below describe intended checks; file presence alone does not prove that they passed.

Size: 2,668 bytes | 41 text lines

Code entry points: DeliveryTests (L9); DeliveryTests.test_canonical_delivery_amount_recipient_network_and_reorg (L10)

Direct imports: unittest, copy, types, web3.datastructures, hexbytes, services.payments.token_delivery

Declared checks: DeliveryTests.test_canonical_delivery_amount_recipient_network_and_reorg

backend/tests/test_transak.py

Test or diagnostic coverage for transak. The extracted cases below describe intended checks; file presence alone does not prove that they passed.

Size: 10,560 bytes | 191 text lines

Code entry points: TransakTests (L15); TransakTests.asyncSetUp (L16); TransakTests.asyncTearDown (L73); TransakTests.create (L78); TransakTests.test_concurrent_sessions_share_encrypted_token (L81); TransakTests.test_provider_errors_redacted_and_redirect_rejected (L93); TransakTests.test_invalid_expiry_and_ip_fail_closed (L102); TransakTests.test_verified_orders_persist_and_follow_status_without_entitlement (L110); TransakTests.test_wrong_wallet_route_identity_or_amount_never_saves (L129); TransakTests.test_payout_session_uses_sell_and_no_forced_blockchain (L141); TransakTests.test_terminal_regression_and_late_capture_require_review (L149); TransakTests.test_completed_amount_changes_do_not_replace_evidence (L165); TransakTests.test_background_reconciliation_claims_once_and_preserves_outages (L174)

Direct imports: asyncio, time, unittest, uuid, unittest.mock, httpx, dotenv, services, services.payments

Declared checks: TransakTests.test_concurrent_sessions_share_encrypted_token; TransakTests.test_provider_errors_redacted_and_redirect_rejected; TransakTests.test_invalid_expiry_and_ip_fail_closed; TransakTests.test_verified_orders_persist_and_follow_status_without_entitlement; TransakTests.test_wrong_wallet_route_identity_or_amount_never_saves; TransakTests.test_payout_session_uses_sell_and_no_forced_blockchain; TransakTests.test_terminal_regression_and_late_capture_require_review; TransakTests.test_completed_amount_changes_do_not_replace_evidence; TransakTests.test_background_reconciliation_claims_once_and_preserves_outages

backend/tests/test_transak_discovery.py

Test or diagnostic coverage for transak discovery. The extracted cases below describe intended checks; file presence alone does not prove that they passed.

Size: 2,235 bytes | 29 text lines

Code entry points: DiscoveryTests (L7); DiscoveryTests.test_runtime_methods_quotes_fees_and_fail_closed_routes (L8)

Direct imports: unittest, unittest.mock, httpx, services.payments

Declared checks: DiscoveryTests.test_runtime_methods_quotes_fees_and_fail_closed_routes

backend/tests/test_vqc.py

Test or diagnostic coverage for vqc. The extracted cases below describe intended checks; file presence alone does not prove that they passed.

Size: 1,240 bytes | 24 text lines

Code entry points: VQCTests (L6); VQCTests.test_ndcg_perfect_zero_and_reverse (L7); VQCTests.test_circuit_expectations_and_training_dimensions (L13)

Direct imports: unittest, numpy, research.vqc_embeddings

Declared checks: VQCTests.test_ndcg_perfect_zero_and_reverse; VQCTests.test_circuit_expectations_and_training_dimensions

backend/tests/test_week2.py

Test or diagnostic coverage for week2. The extracted cases below describe intended checks; file presence alone does not prove that they passed.

Size: 2,950 bytes | 102 text lines

Code entry points: test_text_extractor_txt (L26); test_blake3_hashing (L35); test_fraud_detector (L49); test_llm_enrichment (L72)

Direct imports: pytest, services.text_extractor, services.llm_enrichment, services.quantum_hash, services.fraud_detector

Declared checks: test_text_extractor_txt; test_blake3_hashing; test_fraud_detector; test_llm_enrichment

backend/verify_transak_coverage.py

Historical staging route/quote discovery evidence capture; offers and quotes can expire.

Size: 1,418 bytes | 28 text lines

Code entry points: main (L10)

Direct imports: asyncio, json, pathlib, dotenv, services.payments.transak_discovery

backend/verify_transak_token.py

Historical provider test-token chain verification, not verification of the current MockUSDC purchase.

Size: 3,417 bytes | 70 text lines

Code entry points: abi (L15); verify (L21)

Direct imports: json, os, datetime, pathlib, dotenv, web3, web3.middleware

Directory: contracts

contracts /.agents/skills/hardhat-toolbox-viem/SKILL.md

Legacy/scaffold duplicate under a directory whose name ends in a space. Do not confuse this with the active contracts/ project. Inspect before use; not the current deployment entry point.

Size: 4,455 bytes | 126 text lines

Document sections: Hardhat toolbox: viem; `viem`: clients and contract interaction; `viem.assertions`: Ethereum-specific assertions

contracts /.agents/skills/hardhat/SKILL.md

Legacy/scaffold duplicate under a directory whose name ends in a space. Do not confuse this with the active contracts/ project. Inspect before use; not the current deployment entry point.

Size: 7,072 bytes | 152 text lines

Document sections: Hardhat 3; Test organization; Solidity tests; TypeScript tests and the network connection; `networkHelpers`: EVM state manipulation; Typechecking

contracts /.claude/skills/hardhat-toolbox-viem/SKILL.md

Legacy/scaffold duplicate under a directory whose name ends in a space. Do not confuse this with the active contracts/ project. Inspect before use; not the current deployment entry point.

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Document sections: Hardhat toolbox: viem; `viem`: clients and contract interaction; `viem.assertions`: Ethereum-specific assertions

contracts /.claude/skills/hardhat/SKILL.md

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Size: 7,072 bytes | 152 text lines

Document sections: Hardhat 3; Test organization; Solidity tests; TypeScript tests and the network connection; `networkHelpers`: EVM state manipulation; Typechecking

contracts /.gitignore

Legacy/scaffold duplicate under a directory whose name ends in a space. Do not confuse this with the active contracts/ project. Inspect before use; not the current deployment entry point.

Size: 368 bytes | 31 text lines

contracts /AGENTS.md

Legacy/scaffold duplicate under a directory whose name ends in a space. Do not confuse this with the active contracts/ project. Inspect before use; not the current deployment entry point.

Size: 927 bytes | 20 text lines

Document sections: Hardhat + viem project; Project layout; Working in this project; Docs

contracts /CLAUDE.md

Legacy/scaffold duplicate under a directory whose name ends in a space. Do not confuse this with the active contracts/ project. Inspect before use; not the current deployment entry point.

Size: 11 bytes | 1 text lines

contracts /contracts/Counter.sol

Legacy/scaffold duplicate under a directory whose name ends in a space. Do not confuse this with the active contracts/ project. Counter demonstration code.

Size: 332 bytes | 19 text lines

Code entry points: Counter (L4); inc (L9); incBy (L14)

Events: Increment

contracts /contracts/Counter.t.sol

Legacy/scaffold duplicate under a directory whose name ends in a space. Do not confuse this with the active contracts/ project. Counter demonstration code.

Size: 645 bytes | 29 text lines

Code entry points: CounterTest (L7); setUp (L10); test_InitialValue (L14); testFuzz_Inc (L18); test_IncByZero (L25)

Direct imports: ./Counter.sol, forge-std/Test.sol

contracts /hardhat.config.ts

Legacy/scaffold duplicate under a directory whose name ends in a space. Do not confuse this with the active contracts/ project. Inspect before use; not the current deployment entry point.

Size: 827 bytes | 38 text lines

Direct imports: @nomicfoundation/hardhat-toolbox-viem, hardhat/config

contracts /ignition/modules/Counter.ts

Legacy/scaffold duplicate under a directory whose name ends in a space. Do not confuse this with the active contracts/ project. Counter demonstration code.

Size: 230 bytes | 9 text lines

Direct imports: @nomicfoundation/hardhat-ignition/modules

contracts /package.json

Legacy/scaffold duplicate under a directory whose name ends in a space. Do not confuse this with the active contracts/ project. Inspect before use; not the current deployment entry point.

Size: 68 bytes | 5 text lines

Top-level fields: name, version, type

contracts /README.md

Legacy/scaffold duplicate under a directory whose name ends in a space. Do not confuse this with the active contracts/ project. Inspect before use; not the current deployment entry point.

Size: 2,191 bytes | 57 text lines

Document sections: Sample Hardhat 3 Project (`node:test` and `viem`); Project Overview; Usage; Running Tests; Make a deployment to Sepolia

contracts /scripts/send-op-tx.ts

Legacy/scaffold duplicate under a directory whose name ends in a space. Do not confuse this with the active contracts/ project. Inspect before use; not the current deployment entry point.

Size: 804 bytes | 31 text lines

Direct imports: hardhat

contracts /test/Counter.ts

Legacy/scaffold duplicate under a directory whose name ends in a space. Do not confuse this with the active contracts/ project. Counter demonstration code.

Size: 1,300 bytes | 46 text lines

Direct imports: node:assert/strict, node:test, hardhat

Declared checks: Should emit the Increment event when calling the inc() function; The sum of the Increment events should match the current value

contracts /tsconfig.json

Legacy/scaffold duplicate under a directory whose name ends in a space. Do not confuse this with the active contracts/ project. Inspect before use; not the current deployment entry point.

Size: 206 bytes | 11 text lines

Top-level fields: compilerOptions

Directory: contracts

contracts/.agents/skills/hardhat-toolbox-viem/SKILL.md

Developer/agent workflow instructions and tooling guidance; not executed as an application feature.

Size: 4,455 bytes | 126 text lines

Document sections: Hardhat toolbox: viem; `viem`: clients and contract interaction; `viem.assertions`: Ethereum-specific assertions

contracts/.agents/skills/hardhat/SKILL.md

Developer/agent workflow instructions and tooling guidance; not executed as an application feature.

Size: 7,072 bytes | 152 text lines

Document sections: Hardhat 3; Test organization; Solidity tests; TypeScript tests and the network connection; `networkHelpers`: EVM state manipulation; Typechecking

contracts/.claude/skills/hardhat-toolbox-viem/SKILL.md

Developer/agent workflow instructions and tooling guidance; not executed as an application feature.

Size: 4,455 bytes | 126 text lines

Document sections: Hardhat toolbox: viem; `viem`: clients and contract interaction; `viem.assertions`: Ethereum-specific assertions

contracts/.claude/skills/hardhat/SKILL.md

Developer/agent workflow instructions and tooling guidance; not executed as an application feature.

Size: 7,072 bytes | 152 text lines

Document sections: Hardhat 3; Test organization; Solidity tests; TypeScript tests and the network connection; `networkHelpers`: EVM state manipulation; Typechecking

contracts/.gitignore

Repository/tooling metadata controlling ignores, linting or project behavior; not user-facing application logic.

Size: 368 bytes | 31 text lines

contracts/AGENTS.md

Developer/agent workflow instructions and tooling guidance; not executed as an application feature.

Size: 927 bytes | 20 text lines

Document sections: Hardhat + viem project; Project layout; Working in this project; Docs

contracts/cents-deployment.json

Current Amoy successor deployment receipts, original-state snapshot and wiring verification.

Size: 3,807 bytes | 117 text lines

Top-level fields: status, chain_id, deployer, legacy_registry, legacy_marketplace, legacy_splitter, token, legacy_token_limit, snapshots, contracts, wiring_transaction

contracts/CLAUDE.md

Developer/agent workflow instructions and tooling guidance; not executed as an application feature.

Size: 11 bytes | 1 text lines

contracts/contracts/interfaces/IKnowledgeAsset.sol

Solidity interface defining the cross-contract callable surface and shared types; implementation lives in the corresponding contract.

Size: 1,406 bytes | 68 text lines

Code entry points: IKnowledgeAsset (L4); mint (L23); getAsset (L31); ownerOf (L35); getApproved (L39); isApprovedForAll (L43); safeTransferFrom (L48); updatePrice (L54); deactivate (L59); totalMinted (L63)

contracts/contracts/interfaces/IMarketplace.sol

Solidity interface defining the cross-contract callable surface and shared types; implementation lives in the corresponding contract.

Size: 661 bytes | 34 text lines

Code entry points: IMarketplace (L4); listAsset (L22); purchaseAsset (L27); cancelListing (L31)

Events: AssetListed, AssetPurchased, ListingCancelled

contracts/contracts/interfaces/IRoyaltySplitter.sol

Solidity interface defining the cross-contract callable surface and shared types; implementation lives in the corresponding contract.

Size: 2,524 bytes | 105 text lines

Code entry points: IRoyaltySplitter (L4); calculatePlatformFee (L48); calculateRoyaltyPool (L60); distributeRoyalty (L67); updatePlatformFee (L73); updateTreasury (L78); updateMarketplace (L85); treasury (L90); platformFeeBps (L95); marketplace (L101)

Events: RoyaltyDistributed, RecipientPaid, PlatformFeePaid, PlatformFeeUpdated, TreasuryUpdated, SellerPaid, MarketplaceUpdated

contracts/contracts/KnowledgeAsset.sol

ERC-721 knowledge registration: content/metadata CIDs, creator, price, license, quality and active status. Minting registers content; it is not an asset purchase.

Size: 11,081 bytes | 321 text lines

Code entry points: KnowledgeAsset (L28); mint (L140); getAsset (L201); updatePrice (L217); deactivate (L242); _beforeTokenTransfer (L270); totalMinted (L302); tokenURI (L312)

Events: AssetMinted, AssetDeactivated, PriceUpdated

Custom errors: EmptyIPFSCID, EmptyMetadataCID, PriceTooLow, PriceTooHigh, QualityScoreTooHigh, AssetDoesNotExist, NotAssetOwner, AssetInactive, AlreadyInactive, NonTransferable

contracts/contracts/KnowledgeAssetSuccessor.sol

Successor registry reads the fixed original prefix 1-6 and creates new tokens from 7 onward. Preserves original NFTs without copying their ownership.

Size: 1,623 bytes | 42 text lines

Code entry points: KnowledgeAssetSuccessor (L8); _ownerOf (L21); balanceOf (L26); getAsset (L33); tokenURI (L38)

contracts/contracts/Marketplace.sol

Fixed-price listings, entitlement purchases, funded purchases, auctions, bids, finalization and refunds. Payment token is immutable; buyAsset transfers ERC-20 tokens and grants access, not NFT ownership.

Size: 30,693 bytes | 956 text lines

Code entry points: Marketplace (L38); hasPurchased (L221); listFixedPrice (L321); cancelListing (L400); buyAsset (L430); buyAssetFor (L443); _buyAsset (L458); createAuction (L525); placeBid (L622); withdrawRefund (L694); endAuction (L719); pause (L794); unpause (L802); updateRoyaltySplitter (L816); isListed (L833); listingExists (L851); isAuctionActive (L866); auctionExists (L884); getAuctionBids (L899); getListingRecipients (L914); getAuctionRecipients (L925); totalListingsCreated (L940)

Code entry points: totalAuctionsCreated (L949)

Events: AssetListed, ListingCancelled, PurchaseCompleted, FundedPurchase, AuctionCreated, BidPlaced, AuctionEnded, RefundWithdrawn

Custom errors: InvalidKnowledgeAsset, InvalidRoyaltySplitter, InvalidUSDC, InvalidPrice, NotNFTOwner, MarketplaceNotApproved, AlreadyListed, NoRecipients, InvalidRecipient, InvalidShare, TooManyRecipients, RoyaltyExceedsMaximum, ListingInactive, NotSeller, InvalidSaleType, SellerCannotBuyOwnAsset, SellerNoLongerOwnsNFT, InvalidReservePrice, InvalidAuctionDuration, AuctionAlreadyExists, AuctionInactive, AuctionAlreadyFinalized, AuctionNotStarted, AuctionHasEnded, SellerCannotBid, BelowReservePrice, BidTooLow, AuctionStillRunning, NoRefundOwed, InvalidBuyer, InvalidPaymentReference, PaymentAlreadySettled

contracts/contracts/MockERC20.sol

Test ERC-20 with configured decimals and externally callable mint. Creates test currency; never treat its unrestricted issuance as a production stablecoin.

Size: 672 bytes | 36 text lines

Code entry points: MockERC20 (L6); mint (L19); decimals (L28)

contracts/contracts/PaymentReceiptRegistry.sol

Optional restricted receipt-recording contract. Records evidence identifiers without replacing marketplace entitlement rules.

Size: 3,487 bytes | 74 text lines

Code entry points: PaymentReceiptRegistry (L7); recordPayment (L47); recordReversal (L65)

Events: PaymentRecorded, PaymentReversed

contracts/contracts/RoyaltySplitter.sol

Restricted marketplace caller distributes token proceeds to treasury, royalty recipients and seller with validated basis-point limits.

Size: 14,929 bytes | 460 text lines

Code entry points: RoyaltySplitter (L29); calculatePlatformFee (L144); calculateRoyaltyPool (L166); updateTreasury (L185); updatePlatformFee (L209); updateMarketplace (L236); _validateRecipients (L268); distributeRoyalty (L354)

Custom errors: InvalidTreasury, InvalidUSDC, InvalidPlatformFee, InvalidRecipient, InvalidRoyaltyShare, InvalidAmount, TooManyRecipients, DuplicateRecipient, InvalidSeller, SellerCannotBeRecipient, RoyaltyExceedsMaximum, InvalidMarketplace, NotMarketplace

contracts/deployment.json

Earlier deployment record; compare with the current cents manifest before using addresses.

Size: 318 bytes | 7 text lines

Top-level fields: deployer, MockUSDC, KnowledgeAsset, RoyaltySplitter, Marketplace

contracts/hardhat.compiler-review.config.ts

Build/runtime/tooling configuration for hardhat.compiler-review.config. Review this alongside its package manifest; secrets are deliberately excluded.

Size: 366 bytes | 11 text lines

Direct imports: ./hardhat.config.js

contracts/hardhat.config.ts

Build/runtime/tooling configuration for hardhat.config. Review this alongside its package manifest; secrets are deliberately excluded.

Size: 681 bytes | 35 text lines

Direct imports: hardhat/config, @nomicfoundation/hardhat-toolbox-mocha-ethers, dotenv

contracts/ignition/modules/Counter.ts

Hardhat starter counter example/scaffold, not the DKM marketplace implementation.

Size: 230 bytes | 9 text lines

Direct imports: @nomicfoundation/hardhat-ignition/modules

contracts/package-lock.json

Dependency lockfile: exact resolved dependency graph and integrity metadata. Used for reproducible installation; not application business logic.

Size: 154,276 bytes | 4,382 text lines

Top-level fields: name, version, lockfileVersion, requires, packages

contracts/package.json

Node package manifest: scripts and direct dependencies. Version ranges are not necessarily the versions resolved in the lockfile.

Size: 749 bytes | 30 text lines

Top-level fields: name, version, description, main, scripts, keywords, type, devDependencies, dependencies, allowScripts

contracts/pnpm-lock.yaml

Dependency lockfile: exact resolved dependency graph and integrity metadata. Used for reproducible installation; not application business logic.

Size: 85,774 bytes | 2,537 text lines

contracts/pnpm-workspace.yaml

Build/runtime/tooling configuration for pnpm-workspace. Review this alongside its package manifest; secrets are deliberately excluded.

Size: 50 bytes | 2 text lines

contracts/README.md

Project documentation for README. Treat dated plans and old status notes as historical unless confirmed by current code/deployment.

Size: 2,191 bytes | 57 text lines

Document sections: Sample Hardhat 3 Project (`node:test` and `viem`); Project Overview; Usage; Running Tests; Make a deployment to Sepolia

contracts/scripts/check-balance.ts

Read-only native POL balance diagnostic for the configured signer; not an ERC-20 balance or faucet.

Size: 435 bytes | 19 text lines

Code entry points: main (L3)

Direct imports: hardhat

contracts/scripts/checkAsset.ts

Asset contract/state inspection utility.

Size: 1,138 bytes | 47 text lines

Code entry points: main (L3)

Direct imports: hardhat

contracts/scripts/checkCode.ts

Deployment bytecode-presence diagnostic; bytecode presence alone is not full configuration validation.

Size: 484 bytes | 19 text lines

Code entry points: main (L3)

Direct imports: hardhat

contracts/scripts/decode-error.ts

Contract error decoding helper for revert diagnostics.

Size: 846 bytes | 34 text lines

Code entry points: main (L3)

Direct imports: hardhat

contracts/scripts/deploy-cents.mjs

Successor/market/splitter transition script and verification journal for the live one-cent Amoy deployment. Running it is a deployment action, not a harmless diagnostic.

Size: 5,945 bytes | 72 text lines

Code entry points: main (L11); gasLimit (L39); budget (L42); save (L47); gasLimit (L58)

Direct imports: node:fs, dotenv, ethers

contracts/scripts/deploy-knowledge-asset.ts

Knowledge registry deployment utility; a new address does not migrate old entitlements automatically.

Size: 1,483 bytes | 63 text lines

Code entry points: main (L4)

Direct imports: hardhat, fs

contracts/scripts/deploy-trnsk-sandbox.ts

Historical alternative-token sandbox deployment path; does not describe the active MockUSDC market.

Size: 5,873 bytes | 76 text lines

Code entry points: main (L13); symbol (L23); decimals (L23); missing (L27); record (L51); save (L49)

Direct imports: hardhat, node:fs

contracts/scripts/deploy.ts

Deployment entry point for the earlier contract suite; verify target network and existing-state preservation before running.

Size: 3,670 bytes | 137 text lines

Code entry points: main (L4)

Direct imports: hardhat, fs

contracts/scripts/New Text Document.txt

Auxiliary text file; inspect its actual content before treating it as an executable project entry point.

Size: 0 bytes | binary/unreadable as UTF-8

contracts/scripts/send-op-tx.ts

Operational transaction helper. Review its exact payload and signer before running; not a generic read-only command.

Size: 804 bytes | 31 text lines

Direct imports: hardhat

contracts/scripts/test-list-empty-royalties.ts

Supporting test list empty royalties module. Its directly extracted symbols and references below identify its concrete scope.

Size: 1,287 bytes | 60 text lines

Code entry points: main (L4)

Direct imports: hardhat, ../deployment.json

contracts/scripts/verify-deployment.ts

Checks deployed contract configuration and wiring against expected values.

Size: 1,801 bytes | 76 text lines

Code entry points: main (L4)

Direct imports: hardhat, ../deployment.json

contracts/test/FundedPurchase.test.ts

Test or diagnostic coverage for FundedPurchase. The extracted cases below describe intended checks; file presence alone does not prove that they passed.

Size: 3,911 bytes | 55 text lines

Code entry points: fixture (L6)

Direct imports: chai, hardhat

Declared checks: pays seller and fee, credits only the buyer, and rejects replay after relisting; cannot grant a license without funds and allowance; a failed attempt leaves the reference reusable; rejects wrong price, invalid buyer, seller as buyer, missing reference, and paused purchases

contracts/test/Integration.test.ts

Test or diagnostic coverage for Integration. The extracted cases below describe intended checks; file presence alone does not prove that they passed.

Size: 23,020 bytes | 1,073 text lines

Code entry points: royaltyRecipients (L22); createAuction (L510)

Direct imports: chai, hardhat

Declared checks: completes an entire NFT sale; distributes payment correctly after purchase; updates marketplace state after purchase; emits PurchaseCompleted in integration flow; completes an entire auction workflow; credits, but does not auto-pay, the outbid bidder; stores complete bid history; emits AuctionEnded event; handles multiple fixed-price sales; supports listing and auction simultaneously; handles multiple auctions; maintains marketplace state correctly

contracts/test/KnowledgeAsset.test.ts

Test or diagnostic coverage for KnowledgeAsset. The extracted cases below describe intended checks; file presence alone does not prove that they passed.

Size: 7,456 bytes | 315 text lines

Direct imports: chai, hardhat

Declared checks: mints NFT and stores all asset data correctly; accepts the one-cent mint and update boundary and rejects zero/below-minimum updates; assigns NFT ownership to seller; emits AssetMinted event; reverts when IPFS CID is empty; reverts when metadata CID is empty; reverts when price is below 0.01 USDC; reverts when price is above 100 USDC; reverts when quality score exceeds 100; reverts when looking up a nonexistent asset; allows seller to update price; prevents non-owner from updating price

Declared checks: prevents updating the price of an inactive asset; allows the owner to deactivate an asset; prevents non-owner from deactivating; prevents deactivating an already-inactive asset; emits AssetDeactivated event; returns an ipfs:// tokenURI pointing at the metadata CID

contracts/test/KnowledgeAssetSuccessor.test.ts

Test or diagnostic coverage for KnowledgeAssetSuccessor. The extracted cases below describe intended checks; file presence alone does not prove that they passed.

Size: 3,288 bytes | 48 text lines

Direct imports: chai, hardhat

Declared checks: preserves legacy reads and licenses, separates later legacy mints, and settles a cent purchase

contracts/test/Marketplace.test.ts

Test or diagnostic coverage for Marketplace. The extracted cases below describe intended checks; file presence alone does not prove that they passed.

Size: 35,418 bytes | 1,878 text lines

Code entry points: mintAsset (L27); approveMarketplace (L37); createListing (L46); createAuction (L64); createFinishedAuction (L95); royaltyRecipients (L150)

Direct imports: chai, hardhat

Declared checks: stores KnowledgeAsset correctly; stores RoyaltySplitter correctly; stores USDC correctly; starts with zero listings; starts with zero auctions; seller can mint an NFT; seller approves marketplace; creates a fixed-price listing; emits AssetListed; rejects non-owner listing; rejects duplicate listing; rejects invalid royalty shares

Declared checks: rejects more than MAX_RECIPIENTS royalty recipients; buyer purchases listed NFT; prevents buyer from reselling purchased NFT; marks listing inactive after purchase; emits PurchaseCompleted; prevents seller from buying own NFT; rejects inactive listing; fails when buyer has no USDC; fails without USDC approval; creates auction successfully; auction exists; stores royalty recipients

Declared checks: emits AuctionCreated; rejects duplicate auction; rejects zero reserve price; rejects invalid duration; rejects non-owner; accepts the first valid bid; emits BidPlaced event; rejects bid below reserve; prevents seller from bidding; accepts a higher bid; credits, but does not auto-pay, the outbid bidder; returns true for buyer after purchase

Declared checks: returns false for wallet that has not purchased; rejects withdrawRefund with nothing owed; rejects lower bid; stores bid history; ends auction successfully; keeps NFT with creator after auction; prevents auction winner from reselling purchased NFT; emits AuctionEnded; cannot end auction twice; cannot end auction before end time; handles auction with no bids; clears auction mapping after settlement

Declared checks: marks auction finalized; owner can pause marketplace; owner can unpause marketplace; rejects non-owner pause; updates royalty splitter; rejects zero royalty splitter; listingExists works; auctionExists works; returns listing recipients; returns auction recipients; returns empty bid history; cannot cancel listing twice

Declared checks: listing inactive after cancel; auction inactive after settlement; keeps bid history after settlement

contracts/test/PaymentReceiptRegistry.test.ts

Test or diagnostic coverage for PaymentReceiptRegistry. The extracted cases below describe intended checks; file presence alone does not prove that they passed.

Size: 6,289 bytes | 88 text lines

Code entry points: fixture (L5); original (L69)

Direct imports: chai, hardhat

Declared checks: records different payment and payout denominations without moving funds or NFTs; rejects a zero initial recorder and makes environment classification immutable; rejects unauthorized attestations and duplicate internal/provider references; validates asset, parties, amounts and evidence before consuming references; records terminal reversal ; allows recorder rotation without allowing the revoked signer to write

contracts/test/RoyaltySplitter.test.ts

Test or diagnostic coverage for RoyaltySplitter. The extracted cases below describe intended checks; file presence alone does not prove that they passed.

Size: 14,865 bytes | 596 text lines

Direct imports: chai, hardhat

Declared checks: stores treasury correctly; stores platform fee correctly; calculates platform fee correctly; calculates royalty pool correctly; allows owner to update treasury; prevents non-owner from updating treasury; rejects zero treasury address; allows owner to update platform fee; prevents non-owner from updating platform fee; rejects platform fee above maximum; emits TreasuryUpdated event; emits PlatformFeeUpdated event

Declared checks: allows owner to update the marketplace address; prevents non-owner from updating the marketplace address; rejects zero marketplace address; emits MarketplaceUpdated event; distributes royalties correctly; emits SellerPaid event; rejects calls from anyone other than the marketplace; rejects zero seller address; rejects zero amount; rejects royalty shares above the maximum; rejects a recipient that is also the seller; rejects duplicate recipients

Declared checks: rejects zero recipient address; rejects too many recipients

contracts/test/TRNSK.test.ts

Test or diagnostic coverage for TRNSK. The extracted cases below describe intended checks; file presence alone does not prove that they passed.

Size: 3,976 bytes | 50 text lines

Direct imports: chai, hardhat

Declared checks: preserves legacy licenses and pays exact TRNSK seller proceeds for funded purchases and auctions

contracts/tsconfig.json

Structured tsconfig configuration/evidence. Top-level structure is listed below; its existence alone is not live-state proof.

Size: 316 bytes | 18 text lines

Top-level fields: compilerOptions, include

Directory: Root documentation and evidence

DEMO_RUNBOOK.md

Historical TRNSK/provider-oriented runbook. Superseded for current wallet-only MockUSDC flow; do not follow its provider steps as current instructions.

Size: 3,762 bytes | 53 text lines

Document sections: Local college demo - TRNSK sandbox

ENGINEERING_PLAN.md

Project documentation for ENGINEERING PLAN. Treat dated plans and old status notes as historical unless confirmed by current code/deployment.

Size: 1,899 bytes | 32 text lines

Document sections: Active TRNSK plan - 2026-10-12; Completed before deployment; Approval-dependent sequence

FEATURE_COVERAGE.md

Project documentation for FEATURE COVERAGE. Treat dated plans and old status notes as historical unless confirmed by current code/deployment.

Size: 8,174 bytes | 104 text lines

Document sections: Backend to frontend coverage; Where to use each feature; Security and setup limits; Verification; Entry flow acceptance (2026-10-01)

FINAL_STATUS.md

Project documentation for FINAL STATUS. Treat dated plans and old status notes as historical unless confirmed by current code/deployment.

Size: 2,709 bytes | 38 text lines

Document sections: DKM status - 2026-10-12

Directory: frontend

frontend/.eslintrc.json

Structured .eslintrc configuration/evidence. Top-level structure is listed below; its existence alone is not live-state proof.

Size: 61 bytes | 3 text lines

Top-level fields: extends

frontend/.gitignore

Repository/tooling metadata controlling ignores, linting or project behavior; not user-facing application logic.

Size: 391 bytes | 36 text lines

frontend/app/account/page.tsx

Account and wallet-related settings/view.

Size: 2,479 bytes | 29 text lines

Code entry points: AccountPage (L9); verify (L16)

Direct imports: next/link, react, wagmi, @/providers/AuthProvider, @/lib/api/client, @/components/wallet/WalletButton

frontend/app/auctions/page.tsx

Auction discovery and interaction page using market contracts.

Size: 2,477 bytes | 15 text lines

Code entry points: AuctionsPage (L8)

Direct imports: next/link, react, wagmi, @/lib/blockchain/contracts, @/components/marketplace/AuctionPanel

frontend/app/backend-test/page.tsx

Development API diagnostic page, not the normal customer workflow.

Size: 115 bytes | 5 text lines

Code entry points: BackendTestPage (L3)

Direct imports: next/navigation

frontend/app/dashboard/page.tsx

Account workspace, uploaded drafts, mint recovery and activity/positions entry points.

Size: 4,637 bytes | 28 text lines

Code entry points: Dashboard (L11); refresh (L17); run (L19); DashboardPage (L28)

Direct imports: next/link, react, @/providers/AuthProvider, @/lib/api/client, @/components/marketplace/AccountActivity, @/components/marketplace/IndexedActivity, @/components/marketplace/ReviewHistory, @/components/marketplace/Positions

frontend/app/favicon.ico

Static visual/font asset, consumed by frontend rendering; no executable business logic.

Size: 25,931 bytes | binary/unreadable as UTF-8

frontend/app/fonts/GeistMonoVF.woff

Static visual/font asset, consumed by frontend rendering; no executable business logic.

Size: 67,864 bytes | binary/unreadable as UTF-8

frontend/app/fonts/GeistVF.woff

Static visual/font asset, consumed by frontend rendering; no executable business logic.

Size: 66,268 bytes | binary/unreadable as UTF-8

frontend/app/funding/page.tsx

Redirects directly to /marketplace. Historical provider funding UI is not the active purchase path.

Size: 114 bytes | 2 text lines

Code entry points: FundingPage (L2)

Direct imports: next/navigation

frontend/app/globals.css

Shared style definitions and visual tokens for the application.

Size: 1,830 bytes | 81 text lines

frontend/app/layout.tsx

Root Next.js layout, global styling and providers; wraps page rendering.

Size: 549 bytes | 24 text lines

Code entry points: RootLayout (L8)

Direct imports: @/providers/Web3Provider, @/components/layout/AppShell, @/providers/AuthProvider

frontend/app/login/page.tsx

Login route using the shared authentication form.

Size: 112 bytes | 2 text lines

Code entry points: LoginPage (L2)

Direct imports: @/components/auth/AuthForm

frontend/app/marketplace/page.tsx

Marketplace asset catalog and asset-card interactions.

Size: 4,651 bytes | 166 text lines

Code entry points: AssetItem (L20); MarketplaceContent (L84); MarketplacePage (L164)

Direct imports: react, next/link, next/navigation, @/lib/blockchain/useKnowledgeAssets, @/components/marketplace/AssetCard

frontend/app/mint/page.tsx

File analysis/upload, price/license entry, wallet mint, receipt confirmation and upload/token linking. Resumes saved uploads and pending transaction hashes.

Size: 10,693 bytes | 92 text lines

Code entry points: MintFlow (L14); run (L48); link (L53); mint (L59); finishMint (L74); MintContent (L90); MintPage (L92)

Direct imports: react, next/link, next/navigation, wagmi, viem, viem/chains, @/lib/blockchain/contracts, @/lib/blockchain/purchaseSafety, @/lib/api/backend, @/lib/api/client, @/providers/AuthProvider

frontend/app/page.tsx

Unauthenticated welcome entry point; app functions become available after authentication.

Size: 1,604 bytes | 17 text lines

Code entry points: Home (L3)

Direct imports: next/link

frontend/app/research/page.tsx

Explains QAOA/VQC and displays retained experiment results and limitations.

Size: 4,977 bytes | 40 text lines

Code entry points: ResearchPage (L9)

Direct imports: react, @/lib/api/client

frontend/app/search/page.tsx

Semantic query/filter UI and useful catalog results linking to published assets.

Size: 5,536 bytes | 37 text lines

Code entry points: SearchPage (L8)

Direct imports: next/link, react, lucide-react, @/lib/api/backend

frontend/app/signup/page.tsx

Account creation route using the shared authentication form.

Size: 120 bytes | 2 text lines

Code entry points: SignupPage (L2)

Direct imports: @/components/auth/AuthForm

frontend/app/status/page.tsx

Service/capability diagnostics; healthy configuration is not proof of a completed purchase.

Size: 2,049 bytes | 15 text lines

Code entry points: StatusPage (L6); refresh (L9)

Direct imports: next/link, react, @/lib/api/client

frontend/components.json

Structured components configuration/evidence. Top-level structure is listed below; its existence alone is not live-state proof.

Size: 534 bytes | 25 text lines

Top-level fields: $schema, style, rsc, tsx, tailwind, iconLibrary, rtl, aliases, menuColor, menuAccent, registries

frontend/components/auth/AuthForm.tsx

Shared signup/login form and Google-method availability handling.

Size: 4,757 bytes | 57 text lines

Code entry points: AuthForm (L8); googleLogin (L24); submit (L31)

Direct imports: next/link, next/navigation, react, @/lib/api/client, @/providers/AuthProvider

frontend/components/layout/AppShell.tsx

Session-aware navigation gate: welcome/login/signup before authentication; dashboard/workspace after authentication. Backend remains the actual authorization boundary.

Size: 1,050 bytes | 21 text lines

Code entry points: AppShell (L7)

Direct imports: react, next/navigation, @/providers/AuthProvider, ./Navbar

frontend/components/layout/Navbar.tsx

Authenticated application navigation.

Size: 1,567 bytes | 14 text lines

Code entry points: Navbar (L6)

Direct imports: next/link, next/navigation, @/components/wallet/WalletButton, @/providers/AuthProvider

frontend/components/marketplace/AccountActivity.tsx

Sanitized account history display.

Size: 3,698 bytes | 26 text lines

Code entry points: AccountActivity (L9); load (L19); display (L11)

Direct imports: next/link, react, viem, @/lib/api/client, @/lib/api/capabilities

frontend/components/marketplace/AssetCard.tsx

Main asset interface: listing/cancel, quotes, token balance/allowance checks, purchase, access and download controls.

Size: 33,655 bytes | 953 text lines

Code entry points: licenseName (L54); formatUSDC (L67); AssetCard (L71); handleViewDetails (L237); handleApprove (L274); handleList (L294); handleBuy (L384); formatPayment (L84); auctionId (L121)

Direct imports: next/link, @/providers/AuthProvider, @/lib/api/client, @/lib/api/quotes, ./PurchaseQuoteSummary, react, viem, @/lib/blockchain/paymentToken, wagmi, @tanstack/react-query, viem/chains, ./AuctionPanel, @/lib/api/capabilities, ./LegacyListing, ./Reviews, @/lib/blockchain/purchaseSafety, @/lib/blockchain/contracts, @/lib/blockchain/useMarketplace, @/lib/blockchain/useKnowledgeAssets

frontend/components/marketplace/AssetFunding.tsx

Supporting AssetFunding module. Its directly extracted symbols and references below identify its concrete scope.

Size: 5,725 bytes | 64 text lines

Code entry points: AssetFunding (L12); run (L26)

Direct imports: react, @/lib/api/client, viem

frontend/components/marketplace/AuctionPanel.tsx

Auction creation, bids, finalization and refund interactions with matching market/token.

Size: 11,837 bytes | 153 text lines

Code entry points: AuctionPanel (L18); transact (L64); send (L74); display (L25)

Direct imports: next/link, @/providers/AuthProvider, @/lib/api/capabilities, react, wagmi, @tanstack/react-query, viem, viem/chains, @/lib/blockchain/contracts, @/lib/blockchain/paymentToken, @/lib/blockchain/purchaseSafety

frontend/components/marketplace/BlockchainTest.tsx

Development chain connection diagnostic component.

Size: 966 bytes | 47 text lines

Code entry points: BlockchainTest (L8)

Direct imports: wagmi, viem/chains, @/lib/blockchain/contracts

frontend/components/marketplace/IndexedActivity.tsx

Indexed blockchain activity with coverage limitations.

Size: 2,861 bytes | 22 text lines

Code entry points: IndexedActivity (L7); refresh (L12); amount (L9); marketAmount (L10)

Direct imports: react, @/lib/api/client, @/lib/api/capabilities, @/lib/blockchain/paymentToken

frontend/components/marketplace/LegacyListing.tsx

Operations for previous-market positions, maintaining original contract identity.

Size: 3,120 bytes | 47 text lines

Code entry points: LegacyListing (L9); cancel (L24); id (L21)

Direct imports: react, wagmi, @tanstack/react-query, @/providers/AuthProvider, @/lib/blockchain/contracts, @/lib/blockchain/purchaseSafety

frontend/components/marketplace/PaymentCapabilities.tsx

Displays verified route availability and prevents unsupported payment promises.

Size: 1,033 bytes | 15 text lines

Code entry points: PaymentCapabilities (L5)

Direct imports: @/lib/api/capabilities

frontend/components/marketplace/PayoutHistory.tsx

Supporting PayoutHistory module. Its directly extracted symbols and references below identify its concrete scope.

Size: 2,817 bytes | 21 text lines

Code entry points: PayoutHistory (L5); load (L7); refresh (L9)

Direct imports: react, @/lib/api/client

frontend/components/marketplace/Positions.tsx

Wallet-specific positions and actions based on backend/on-chain reads.

Size: 2,978 bytes | 21 text lines

Code entry points: Positions (L9); amount (L11)

Direct imports: next/link, react, @/lib/api/client, @/lib/api/capabilities, @/lib/blockchain/paymentToken

frontend/components/marketplace/PurchaseQuoteSummary.tsx

Displays purchase total and seller deductions; keeps conversion/funding separate from asset purchase.

Size: 1,097 bytes | 17 text lines

Code entry points: PurchaseQuoteSummary (L4); amount (L6)

Direct imports: viem, @/lib/api/quotes

frontend/components/marketplace/RecordingHistory.tsx

Supporting RecordingHistory module. Its directly extracted symbols and references below identify its concrete scope.

Size: 1,390 bytes | 14 text lines

Code entry points: RecordingHistory (L5)

Direct imports: react, @/lib/api/client

frontend/components/marketplace/ReviewHistory.tsx

Account review history display.

Size: 1,077 bytes | 10 text lines

Code entry points: ReviewHistory (L6)

Direct imports: next/link, react, @/lib/api/client

frontend/components/marketplace/Reviews.tsx

Read/write review interface with backend entitlement enforcement.

Size: 3,086 bytes | 18 text lines

Code entry points: Reviews (L6)

Direct imports: react, @/lib/api/client

frontend/components/marketplace/SalePayout.tsx

Supporting SalePayout module. Its directly extracted symbols and references below identify its concrete scope.

Size: 4,378 bytes | 34 text lines

Code entry points: SalePayout (L8); act (L21)

Direct imports: react, viem, @/lib/api/client, @/providers/AuthProvider

frontend/components/marketplace/SellerPayout.tsx

Supporting SellerPayout module. Its directly extracted symbols and references below identify its concrete scope.

Size: 3,890 bytes | 55 text lines

Code entry points: SellerPayout (L7); save (L30)

Direct imports: react, @/lib/api/client, @/providers/AuthProvider, ./SalePayout

frontend/components/marketplace/TransakFunding.tsx

Supporting TransakFunding module. Its directly extracted symbols and references below identify its concrete scope.

Size: 2,724 bytes | 38 text lines

Code entry points: TransakFunding (L5); start (L13)

Direct imports: react, @/lib/api/client

frontend/components/marketplace/TransakOrders.tsx

Supporting TransakOrders module. Its directly extracted symbols and references below identify its concrete scope.

Size: 4,554 bytes | 60 text lines

Code entry points: TransakOrders (L12); load (L17); refresh (L22)

Direct imports: react, @/lib/api/client

frontend/components/marketplace/Web2Checkout.tsx

Supporting Web2Checkout module. Its directly extracted symbols and references below identify its concrete scope.

Size: 10,167 bytes | 150 text lines

Code entry points: Window (L22); request (L31); Web2Checkout (L38); checkout (L74); refresh (L106)

Direct imports: @/lib/api/client, @/providers/AuthProvider, next/script, react, @tanstack/react-query, viem

frontend/components/ui/avatar.tsx

Reusable avatar UI primitive. Supplies presentation/interaction behavior, not payment or authorization policy.

Size: 3,038 bytes | 109 text lines

Code entry points: Avatar (L8); AvatarImage (L28); AvatarFallback (L41); AvatarBadge (L57); AvatarGroup (L73); AvatarGroupCount (L86)

Direct imports: react, @base-ui/react/avatar, @/lib/utils

frontend/components/ui/badge.tsx

Reusable badge UI primitive. Supplies presentation/interaction behavior, not payment or authorization policy.

Size: 1,925 bytes | 52 text lines

Code entry points: Badge (L30)

Direct imports: @base-ui/react/merge-props, @base-ui/react/use-render, class-variance-authority, @/lib/utils

frontend/components/ui/button.tsx

Reusable button UI primitive. Supplies presentation/interaction behavior, not payment or authorization policy.

Size: 3,240 bytes | 58 text lines

Code entry points: Button (L43)

Direct imports: @base-ui/react/button, class-variance-authority, @/lib/utils

frontend/components/ui/card.tsx

Reusable card UI primitive. Supplies presentation/interaction behavior, not payment or authorization policy.

Size: 2,630 bytes | 103 text lines

Code entry points: Card (L5); CardHeader (L23); CardTitle (L36); CardDescription (L49); CardAction (L59); CardContent (L72); CardFooter (L82)

Direct imports: react, @/lib/utils

frontend/components/ui/dialog.tsx

Reusable dialog UI primitive. Supplies presentation/interaction behavior, not payment or authorization policy.

Size: 4,075 bytes | 160 text lines

Code entry points: Dialog (L10); DialogTrigger (L14); DialogPortal (L18); DialogClose (L22); DialogOverlay (L26); DialogContent (L42); DialogHeader (L83); DialogFooter (L93); DialogTitle (L120); DialogDescription (L133)

Direct imports: react, @base-ui/react/dialog, @/lib/utils, @/components/ui/button, lucide-react

frontend/components/ui/dropdown-menu.tsx

Reusable dropdown-menu UI primitive. Supplies presentation/interaction behavior, not payment or authorization policy.

Size: 8,736 bytes | 268 text lines

Code entry points: DropdownMenu (L9); DropdownMenuPortal (L13); DropdownMenuTrigger (L17); DropdownMenuContent (L21); DropdownMenuGroup (L52); DropdownMenuLabel (L56); DropdownMenuItem (L76); DropdownMenuSub (L99); DropdownMenuSubTrigger (L103); DropdownMenuSubContent (L127); DropdownMenuCheckboxItem (L148); DropdownMenuRadioGroup (L182); DropdownMenuRadioItem (L191); DropdownMenuSeparator (L223); DropdownMenuShortcut (L236)

Direct imports: react, @base-ui/react/menu, @/lib/utils, lucide-react

frontend/components/ui/input.tsx

Reusable input UI primitive. Supplies presentation/interaction behavior, not payment or authorization policy.

Size: 1,040 bytes | 20 text lines

Code entry points: Input (L6)

Direct imports: react, @base-ui/react/input, @/lib/utils

frontend/components/ui/separator.tsx

Reusable separator UI primitive. Supplies presentation/interaction behavior, not payment or authorization policy.

Size: 545 bytes | 25 text lines

Code entry points: Separator (L7)

Direct imports: @base-ui/react/separator, @/lib/utils

frontend/components/ui/sheet.tsx

Reusable sheet UI primitive. Supplies presentation/interaction behavior, not payment or authorization policy.

Size: 4,433 bytes | 138 text lines

Code entry points: Sheet (L10); SheetTrigger (L14); SheetClose (L18); SheetPortal (L22); SheetOverlay (L26); SheetContent (L39); SheetHeader (L83); SheetFooter (L93); SheetTitle (L103); SheetDescription (L116)

Direct imports: react, @base-ui/react/dialog, @/lib/utils, @/components/ui/button, lucide-react

frontend/components/ui/toast.tsx

Reusable toast UI primitive. Supplies presentation/interaction behavior, not payment or authorization policy.

Size: 7,394 bytes | 234 text lines

Code entry points: ToastProvider (L12); ToastPortal (L16); ToastViewport (L20); Toast (L33); ToastContent (L60); ToastTitle (L73); ToastDescription (L83); ToastAction (L96); ToastClose (L111); ToastIcon (L135); ToastList (L182); Toaster (L200)

Direct imports: react, @base-ui/react/toast, @/lib/utils, @/components/ui/button, lucide-react

frontend/components/wallet/WalletButton.tsx

Wallet connection control; connection alone does not prove account-wallet ownership.

Size: 148 bytes | 7 text lines

Code entry points: WalletButton (L5)

Direct imports: @rainbow-me/rainbowkit

frontend/contracts/abi/KnowledgeAsset.json

Contract ABI/interface data used to encode calls and decode outputs/events. An ABI is not a deployment or evidence of a successful transaction.

Size: 44,165 bytes | 685 text lines

Top-level fields: _format, contractName, sourceName, abi, bytecode, deployedBytecode, linkReferences, deployedLinkReferences, immutableReferences, inputSourceName, buildInfoId

frontend/contracts/abi/Marketplace.json

Contract ABI/interface data used to encode calls and decode outputs/events. An ABI is not a deployment or evidence of a successful transaction.

Size: 67,867 bytes | 1,155 text lines

Top-level fields: _format, contractName, sourceName, abi, bytecode, deployedBytecode, linkReferences, deployedLinkReferences, immutableReferences, inputSourceName, buildInfoId

frontend/contracts/abi/MockERC20.json

Contract ABI/interface data used to encode calls and decode outputs/events. An ABI is not a deployment or evidence of a successful transaction.

Size: 17,792 bytes | 330 text lines

Top-level fields: _format, contractName, sourceName, abi, bytecode, deployedBytecode, linkReferences, deployedLinkReferences, immutableReferences, inputSourceName, buildInfoId

frontend/contracts/abi/RoyaltySplitter.json

Contract ABI/interface data used to encode calls and decode outputs/events. An ABI is not a deployment or evidence of a successful transaction.

Size: 28,380 bytes | 514 text lines

Top-level fields: _format, contractName, sourceName, abi, bytecode, deployedBytecode, linkReferences, deployedLinkReferences, immutableReferences, inputSourceName, buildInfoId

frontend/contracts/addresses/index.ts

Current and legacy public contract addresses and registry identity helpers.

Size: 510 bytes | 10 text lines

frontend/lib/api/backend.ts

Typed backend request/response helpers for upload, search, download and other app operations.

Size: 1,771 bytes | 56 text lines

Code entry points: analyzeAsset (L45); uploadAssetToIPFS (L49); searchAssets (L53); checkBackendHealth (L56)

Direct imports: ./client

frontend/lib/api/capabilities.ts

Polling query for chain-verified market profiles and route gating.

Size: 1,313 bytes | 24 text lines

Code entry points: usePaymentCapabilities (L16); marketRouteAvailable (L20)

Direct imports: @tanstack/react-query, ./client

frontend/lib/api/client.ts

Shared authenticated backend request handling and normalized errors.

Size: 1,401 bytes | 26 text lines

Code entry points: downloadAsset (L3); api (L15)

frontend/lib/api/quotes.ts

Validates accepted terms against buyer, chain, token, amount, asset and listing before signing.

Size: 2,006 bytes | 26 text lines

Code entry points: acceptPurchaseQuote (L11)

Direct imports: ./client

frontend/lib/blockchain/amoyFees.ts

Checks chain ID and computes EIP-1559 fee bounds with the observed 25-gwei priority floor.

Size: 989 bytes | 20 text lines

Code entry points: amoyFees (L6)

Direct imports: viem

frontend/lib/blockchain/contracts.ts

Typed contract address/ABI bindings for wallet/RPC operations.

Size: 1,414 bytes | 34 text lines

Code entry points: marketsForToken (L34)

Direct imports: viem, @/contracts/abi/KnowledgeAsset.json, @/contracts/abi/Marketplace.json, @/contracts/abi/RoyaltySplitter.json, @/contracts/abi/MockERC20.json, @/contracts/addresses

frontend/lib/blockchain/paymentToken.ts

Verified market token binding, decimal formatting and exact unit parsing.

Size: 1,009 bytes | 19 text lines

Code entry points: paymentToken (L4); formatTokenAmount (L8); paymentAmount (L12)

Direct imports: viem, @/lib/api/capabilities

frontend/lib/blockchain/purchaseSafety.ts

Shared listing/ownership/entitlement, receipt and gas preflight protections reused across wallet writes.

Size: 4,537 bytes | 62 text lines

Code entry points: PurchaseCheckError (L7); checkUnlisted (L9); checkListing (L17); checkGas (L38); confirmedReceipt (L51); gas (L44)

Direct imports: ./amoyFees.ts, viem

frontend/lib/blockchain/useKnowledgeAssets.ts

Registry read hooks for asset metadata, ownership and totals on Amoy.

Size: 1,736 bytes | 67 text lines

Code entry points: useTotalMinted (L10); useKnowledgeAsset (L19); useAssetOwner (L29); useIsListed (L43); useApproveMarketplace (L57); useListFixedPrice (L65)

Direct imports: wagmi, @/lib/blockchain/contracts

frontend/lib/blockchain/useMarketplace.ts

React hooks for market reads/writes and token approval; callers still need fresh state and receipt verification.

Size: 4,918 bytes | 181 text lines

Code entry points: useIsListed (L21); useTokenToListing (L34); useListing (L47); useMarketplaceWrite (L67); approveNFT (L74); listFixedPrice (L90); buyAsset (L114); useUSDCWrite (L149); approveMarketplace (L153)

Direct imports: wagmi, @/lib/blockchain/purchaseSafety, viem/chains, @/lib/blockchain/contracts

frontend/lib/blockchain/wagmi.ts

Wallet connector and supported-chain configuration.

Size: 308 bytes | 12 text lines

Direct imports: @rainbow-me/rainbowkit, wagmi/chains

frontend/lib/utils.ts

Supporting utils module. Its directly extracted symbols and references below identify its concrete scope.

Size: 166 bytes | 6 text lines

Code entry points: cn (L4)

Direct imports: clsx, tailwind-merge

frontend/next.config.mjs

Build/runtime/tooling configuration for next.config. Review this alongside its package manifest; secrets are deliberately excluded.

Size: 269 bytes | 13 text lines

frontend/package-lock.json

Dependency lockfile: exact resolved dependency graph and integrity metadata. Used for reproducible installation; not application business logic.

Size: 612,286 bytes | 17,083 text lines

Top-level fields: name, version, lockfileVersion, requires, packages

frontend/package.json

Node package manifest: scripts and direct dependencies. Version ranges are not necessarily the versions resolved in the lockfile.

Size: 1,225 bytes | 47 text lines

Top-level fields: name, version, private, scripts, dependencies, devDependencies

frontend/postcss.config.mjs

Build/runtime/tooling configuration for postcss.config. Review this alongside its package manifest; secrets are deliberately excluded.

Size: 135 bytes | 8 text lines

frontend/providers/AuthProvider.tsx

Shared account/session state and authentication actions.

Size: 2,538 bytes | 43 text lines

Code entry points: AuthProvider (L10); logout (L23); useVerifiedWallet (L31); AccountGate (L36); check (L19); useAuth (L30)

Direct imports: next/link, react, wagmi, @/lib/api/client

frontend/providers/Web3Provider.tsx

Wagmi/RainbowKit/query providers for wallet and chain interactions.

Size: 713 bytes | 37 text lines

Code entry points: Web3Provider (L19)

Direct imports: react, @tanstack/react-query, @rainbow-me/rainbowkit, wagmi, @/lib/blockchain/wagmi

frontend/README.md

Project documentation for README. Treat dated plans and old status notes as historical unless confirmed by current code/deployment.

Size: 1,450 bytes | 36 text lines

Document sections: Getting Started; or; or; or; Learn More; Deploy on Vercel

frontend/tailwind.config.ts

Build/runtime/tooling configuration for tailwind.config. Review this alongside its package manifest; secrets are deliberately excluded.

Size: 1,695 bytes | 74 text lines

Direct imports: tailwindcss

frontend/tests/amoyFees.test.mjs

Test or diagnostic coverage for amoyFees. The extracted cases below describe intended checks; file presence alone does not prove that they passed.

Size: 1,221 bytes | 23 text lines

Code entry points: client (L6)

Direct imports: node:assert/strict, node:test, ../lib/blockchain/amoyFees.ts

Declared checks: Amoy mint replaces the rejected 1 gwei tip and raises the fee cap; higher network estimates remain intact; wrong network, absent base fee and RPC failures block signing

frontend/tests/assetFunding.test.mjs

Test or diagnostic coverage for assetFunding. The extracted cases below describe intended checks; file presence alone does not prove that they passed.

Size: 2,943 bytes | 32 text lines

Code entry points: find (L24); render (L23)

Direct imports: node:assert/strict, node:test, node:fs, node:module, node:vm, typescript

Declared checks: funding review precedes hosted checkout and only verified bound delivery reports funding

frontend/tests/capabilities.test.mjs

Test or diagnostic coverage for capabilities. The extracted cases below describe intended checks; file presence alone does not prove that they passed.

Size: 1,500 bytes | 26 text lines

Direct imports: node:assert/strict, node:test, node:fs, node:vm, typescript

Declared checks: payment availability fails closed on unknown route, wrong token/network and paused purchases

frontend/tests/checkoutReview.test.mjs

Test or diagnostic coverage for checkoutReview. The extracted cases below describe intended checks; file presence alone does not prove that they passed.

Size: 4,301 bytes | 78 text lines

Code entry points: render (L40); button (L41)

Direct imports: node:assert/strict, node:test, node:fs, node:module, node:vm, typescript

Declared checks: checkout shows the persisted total before opening Razorpay and reuses the order

frontend/tests/legacyAuction.test.mjs

Test or diagnostic coverage for legacyAuction. The extracted cases below describe intended checks; file presence alone does not prove that they passed.

Size: 3,785 bytes | 50 text lines

Code entry points: find (L35); render (L34)

Direct imports: node:assert/strict, node:test, node:fs, node:module, node:vm, typescript

Declared checks: old auction reads, approval, bid, finalization and refund target the selected market

frontend/tests/legacyListing.test.mjs

Test or diagnostic coverage for legacyListing. The extracted cases below describe intended checks; file presence alone does not prove that they passed.

Size: 2,484 bytes | 36 text lines

Code entry points: render (L27)

Direct imports: node:assert/strict, node:test, node:fs, node:module, node:vm, typescript

Declared checks: legacy listing cancellation targets the old market and requires verified seller

frontend/tests/mintResume.test.mjs

Test or diagnostic coverage for mintResume. The extracted cases below describe intended checks; file presence alone does not prove that they passed.

Size: 1,187 bytes | 24 text lines

Direct imports: node:assert/strict, node:test, node:fs, node:module, node:vm, typescript

Declared checks: mint resume uses the browser query even without server page props

frontend/tests/paymentToken.test.mjs

Test or diagnostic coverage for paymentToken. The extracted cases below describe intended checks; file presence alone does not prove that they passed.

Size: 1,100 bytes | 20 text lines

Direct imports: node:assert/strict, node:test, node:fs, node:module, node:vm, typescript

Declared checks: payment parsing keeps exact six/eighteen decimals and rejects rounded bids

frontend/tests/purchaseQuote.test.mjs

Test or diagnostic coverage for purchaseQuote. The extracted cases below describe intended checks; file presence alone does not prove that they passed.

Size: 2,286 bytes | 30 text lines

Direct imports: node:assert/strict, node:test, node:fs, node:vm, typescript

Declared checks: quote acceptance rejects changed totals, expiry and deployment before wallet execution

frontend/tests/purchaseSafety.test.mjs

Test or diagnostic coverage for purchaseSafety. The extracted cases below describe intended checks; file presence alone does not prove that they passed.

Size: 5,491 bytes | 92 text lines

Code entry points: client (L15); make (L65)

Direct imports: node:assert/strict, node:test, node:fs, viem, ../lib/blockchain/purchaseSafety.ts

Declared checks: listing is blocked by any current or legacy active listing and failed reads; a live ABI-shaped asset remains purchasable and inactive assets are blocked; fresh listing rejects relisting, changed price, cancellation, inactive asset and existing access; gas estimate includes a 20 percent margin and fails before signing; two-block receipt verification rejects reorg, disappearance, revert, shallow depth and replacement; an old marketplace entitlement blocks a duplicate purchase

frontend/tests/registryTransition.test.mjs

Test or diagnostic coverage for registryTransition. The extracted cases below describe intended checks; file presence alone does not prove that they passed.

Size: 1,135 bytes | 23 text lines

Direct imports: node:assert/strict, node:test, node:fs, node:vm, typescript

Declared checks: old marketplace entitlements apply only to the preserved six-token prefix

frontend/tests/sellerPayout.test.mjs

Test or diagnostic coverage for sellerPayout. The extracted cases below describe intended checks; file presence alone does not prove that they passed.

Size: 2,620 bytes | 46 text lines

Code entry points: render (L33); button (L34); payout (L35)

Direct imports: node:assert/strict, node:test, node:fs, node:module, node:vm, typescript

Declared checks: seller preference must be saved before the sale-bound withdrawal flow is enabled

frontend/tests/transakFunding.test.mjs

Test or diagnostic coverage for transakFunding. The extracted cases below describe intended checks; file presence alone does not prove that they passed.

Size: 2,195 bytes | 38 text lines

Code entry points: render (L21); button (L22)

Direct imports: node:assert/strict, node:test, node:fs, node:module, node:vm, typescript

Declared checks: Transak only opens staging and never confirms a marketplace payment

frontend/tests/transakOrders.test.mjs

Test or diagnostic coverage for transakOrders. The extracted cases below describe intended checks; file presence alone does not prove that they passed.

Size: 3,038 bytes | 58 text lines

Code entry points: render (L32); find (L33)

Direct imports: node:assert/strict, node:test, node:fs, node:module, node:vm, typescript

Declared checks: sandbox history verifies through backend, preserves amounts, and reports failures

frontend/tests/walletFees.test.mjs

Test or diagnostic coverage for walletFees. The extracted cases below describe intended checks; file presence alone does not prove that they passed.

Size: 2,309 bytes | 49 text lines

Direct imports: node:assert/strict, node:test, node:fs, node:module, node:vm, typescript

Declared checks: every marketplace hook forwards estimated gas and fees and blocks failed estimates

frontend/tsconfig.json

Structured tsconfig configuration/evidence. Top-level structure is listed below; its existence alone is not live-state proof.

Size: 638 bytes | 28 text lines

Top-level fields: compilerOptions, include, exclude

Directory: Root documentation and evidence

HUMAN_ACTION_REQUIRED.md

Project documentation for HUMAN ACTION REQUIRED. Treat dated plans and old status notes as historical unless confirmed by current code/deployment.

Size: 7,929 bytes | 141 text lines

Document sections: Human actions - TRNSK college sandbox, 2026-10-12; Current gate: approval received; deployer setup required; ACTION REQUIRED: Approve Amoy sandbox deployment; Later: test-wallet funding and local signer setup; Later: hosted checkout and signed acceptance; Google, backup and optional historical RPC

LOCAL_PAYMENT_DEMO.md

Project documentation for LOCAL PAYMENT DEMO. Treat dated plans and old status notes as historical unless confirmed by current code/deployment.

Size: 7,180 bytes | 124 text lines

Document sections: Superseding sandbox route - 2026-10-12; Local payment demo - 2026-10-10; Current acceptance update - 2026-10-11; What was verified; Run locally; Test Transak; Test marketplace separately; Remaining requirements - not completed by staging credentials; Seller withdrawal sandbox

MASTER_TODO.md

Project documentation for MASTER TODO. Treat dated plans and old status notes as historical unless confirmed by current code/deployment.

Size: 1,902 bytes | 31 text lines

Document sections: DKM checklist - 2026-10-12; Implemented and locally tested; Human approval/configuration and subsequent engineering

migration_inventory.json

Structured migration inventory configuration/evidence. Top-level structure is listed below; its existence alone is not live-state proof.

Size: 2,703 bytes | 82 text lines

Top-level fields: captured_at, chain_id, block, block_hash, knowledge_asset, frontend_backend_match, legacy_marketplaces, markets, index_coverage, all_historical_buyers_enumerated, limitation, deployment_authorized

MIGRATION_PLAN.md

Project documentation for MIGRATION PLAN. Treat dated plans and old status notes as historical unless confirmed by current code/deployment.

Size: 3,279 bytes | 63 text lines

Document sections: TRNSK sandbox migration - 2026-10-12; Preserved contracts; Deployment preparation; Activation after approval; Rollback

OWNER_HANDOFF.md

Project documentation for OWNER HANDOFF. Treat dated plans and old status notes as historical unless confirmed by current code/deployment.

Size: 13,692 bytes | 208 text lines

Document sections: DKM handoff: completed work and owner actions; Completed in this continuation; Latest payment owner actions (2026-10-01); Existing Amoy demonstration setup; Deployment work still on the engineering side; Acceptance run once deployment and credentials are ready; Production decisions and access you must supply; Observability and support procedure; Current boundaries; Verification for this handoff; Legacy compatibility code now available (2026-10-09)

PAYMENT_AUTH_VERIFICATION.md

Project documentation for PAYMENT AUTH VERIFICATION. Treat dated plans and old status notes as historical unless confirmed by current code/deployment.

Size: 5,023 bytes | 78 text lines

Document sections: Google login, asset descriptions and payment verification; Google setup; What actually handles INR and USDC; Asset presentation; Checks and limits

PAYMENT_INTEGRATION_OPTIONS.md

Project documentation for PAYMENT INTEGRATION OPTIONS. Treat dated plans and old status notes as historical unless confirmed by current code/deployment.

Size: 5,251 bytes | 73 text lines

Document sections: Future payment options - research only, 17 October 2026; What each route would require; Required implementation, if authorized later; Fees and testing

PAYMENT_RECORDING.md

Project documentation for PAYMENT RECORDING. Treat dated plans and old status notes as historical unless confirmed by current code/deployment.

Size: 4,235 bytes | 73 text lines

Document sections: TRNSK denomination update - 2026-10-12; Audit-only blockchain payment records; Implemented flow; States and reversals; Configuration and operations

PAYMENT_TODO.md

Project documentation for PAYMENT TODO. Treat dated plans and old status notes as historical unless confirmed by current code/deployment.

Size: 7,932 bytes | 124 text lines

Document sections: Payment delivery checklist and owner actions; Agreed requirements and engineering checklist; Owner tasks: do these in order; 1. Obtain Onmeta sandbox access and a written feasibility response; 2. Prepare embedded-wallet access; 3. Supply operational choices; 4. Participate in the acceptance run after engineering readiness; Fee acceptance example (illustrative, not provider pricing); Verification

PRODUCTION_PAYMENTS.md

Project documentation for PRODUCTION PAYMENTS. Treat dated plans and old status notes as historical unless confirmed by current code/deployment.

Size: 14,740 bytes | 227 text lines

Document sections: Production payments and recovery decision; Current direction (2026-10-01); Earlier decision and interim flow; Provider selection evidence and acceptance criteria; Current Amoy payment recovery implementation; Operator procedure; Reservations and timeout boundaries; Production release evidence still required; Existing cryptocurrency funding (Web3)

PROJECT_CONTEXT.md

Chronological continuation record. Newer entries supersede older deployment/UI status; read newest first.

Size: 37,272 bytes | 575 text lines

Document sections: DKM continuation - 2026-09-30; Mint recovery and cancellation - 2026-10-21; LIVE one-cent Amoy transition - 2026-10-21; One-cent contract source minimum - 2026-10-21; Wallet gas preparation - 2026-10-21; UI, search and evidence-based scoring - 2026-10-20; Wallet-only UI and cent pricing - 2026-10-17; Test-wallet funding update - 2026-10-14; Deployment approval received - 2026-10-12; Active TRNSK continuation - 2026-10-12 (pre-deployment checks passed); Current continuation (2026-10-11; supersedes stale blocker notes below); Project rules; Current deployment; Implemented in this branch; Verified; Next work and blockers; Verification commands; Account and feature coverage continuation (additional historical sections follow)

PROVIDER_RECONCILIATION.md

Project documentation for PROVIDER RECONCILIATION. Treat dated plans and old status notes as historical unless confirmed by current code/deployment.

Size: 2,236 bytes | 38 text lines

Document sections: Transak saved-order reconciliation

README.md

Project documentation for README. Treat dated plans and old status notes as historical unless confirmed by current code/deployment.

Size: 37 bytes | 2 text lines

Document sections: capstone project

RESEARCH_PLAN.md

Project documentation for RESEARCH PLAN. Treat dated plans and old status notes as historical unless confirmed by current code/deployment.

Size: 4,245 bytes | 76 text lines

Document sections: Required QAOA / VQC work; QAOA pricing; VQC embeddings; Verification and report; Reproduce and inspect

Directory: research_results

research_results/qaoa_pricing.json

Retained QAOA per-case outcomes, configuration, objective gaps, runtimes and simulator provenance.

Size: 650,633 bytes | 24,608 text lines

Top-level fields: environment, dataset, dataset_seed, price_grid_usdc, platform_fee_bps, royalty_assumption, versions, summary, assets

research_results/QAOA_PRICING_REPORT.md

Project documentation for QAOA PRICING REPORT. Treat dated plans and old status notes as historical unless confirmed by current code/deployment.

Size: 1,829 bytes | 29 text lines

Document sections: QAOA pricing benchmark; Method and limitations

research_results/vqc_embeddings.json

Retained VQC corpus/split/configuration, per-query rankings, metrics, versions and training trace.

Size: 125,126 bytes | 5,979 text lines

Top-level fields: environment, dataset, training, training_document_ids, queries_used_for_training, seed, epochs, qubits, layers, compressed_dimensions, quantum_features, output_dimensions, embedding_model, versions, model_sha256, training_seconds, transform_seconds, loss_trace, summary, documents, queries

research_results/VQC_EMBEDDINGS_REPORT.md

Project documentation for VQC EMBEDDINGS REPORT. Treat dated plans and old status notes as historical unless confirmed by current code/deployment.

Size: 1,827 bytes | 18 text lines

Document sections: VQC embedding benchmark; Method and limits

research_results/vqc_model.npz

Fitted VQC/PCA/decoder arrays; binary experiment artifact, not a production search model.

Size: 84,881 bytes | binary/unreadable as UTF-8

Directory: Root documentation and evidence

restore_drill_evidence.json

Saved local database restore comparison; external IPFS and independent key-backup recovery were not tested.

Size: 1,303 bytes | 67 text lines

Top-level fields: scope, external_ipfs_restore_tested, independent_key_backup_tested, tables

ROADMAP_STATUS.md

Project documentation for ROADMAP STATUS. Treat dated plans and old status notes as historical unless confirmed by current code/deployment.

Size: 16,089 bytes | 200 text lines

Document sections: Superseding TRNSK update - 2026-10-12; DKM roadmap audit - updated 2026-10-11; Work added during this audit; Encrypted storage operations; Verification and external limits; Next execution prerequisites; Slither scan and triage; Production payment preparation update; Funding readiness and handoff verification

sandbox-deployment-addresses.example.json

Structured sandbox deployment addresses.example configuration/evidence. Top-level structure is listed below; its existence alone is not live-state proof.

Size: 232 bytes | 6 text lines

Top-level fields: treasury, admin, settlement, recorder

sandbox-deployment-addresses.json

Structured sandbox deployment addresses configuration/evidence. Top-level structure is listed below; its existence alone is not live-state proof.

Size: 242 bytes | 6 text lines

Top-level fields: treasury, admin, settlement, recorder

SECURITY_NOTES.md

Project documentation for SECURITY NOTES. Treat dated plans and old status notes as historical unless confirmed by current code/deployment.

Size: 5,153 bytes | 86 text lines

Document sections: TRNSK continuation checks - 2026-10-12; Local demo security evidence - 2026-10-11; Compiler and static analysis; Payments and workers; Restore drill; Remaining acceptance boundaries

SLITHER_AUDIT.json

Saved static-analysis findings. Source analysis is not an independent audit or proof that every deployed bytecode version was analyzed.

Size: 9,085 bytes | 129 text lines

Top-level fields: initial, final

TEST_EVIDENCE.md

Recorded test and verification history, including failures and evidence boundaries. Counts belong to separate dated runs.

Size: 10,885 bytes | 158 text lines

Document sections: Latest TRNSK acceptance - 2026-10-12; Verification evidence; Autonomous remaining-work continuation (after b49d5f6); Research and BUY reconciliation milestone (2026-10-11); 2026-10-11 baseline (commit 6612caf); Capability milestone; Fixed-price purchase quote milestone

transak_coverage_evidence.json

Structured transak coverage evidence configuration/evidence. Top-level structure is listed below; its existence alone is not live-state proof.

Size: 2,530 bytes | 108 text lines

Top-level fields: verified, environment, checked_at, fiat, sandbox_route, limitation, buy_quote_probe

transak_token_evidence.json

Structured transak token evidence configuration/evidence. Top-level structure is listed below; its existence alone is not live-state proof.

Size: 635 bytes | 17 text lines

Top-level fields: verified, chain_id, token, symbol, decimals, block, block_hash, code_hash, probe_balance, probe_allowance, approve_zero_eth_call, transfer_zero_eth_call, broadcast, checked_at, limitation

trnsk-deployment-plan.json

Structured trnsk deployment plan configuration/evidence. Top-level structure is listed below; its existence alone is not live-state proof.

Size: 1,393 bytes | 30 text lines

Top-level fields: chain_id, sandbox, verified_block, existing_code_hashes, preserved, token, roles, platform_fee_bps

trnsk-deployment-preflight.json

Structured trnsk deployment preflight configuration/evidence. Top-level structure is listed below; its existence alone is not live-state proof.

Size: 526 bytes | 15 text lines

Top-level fields: chain_id, block, admin, balance_test_POL, gas_price_wei, deployment_gas, wiring_gas_budget, deployment_budget_test_POL, budget_note

trnsk-wallet-funding.json

Structured trnsk wallet funding configuration/evidence. Top-level structure is listed below; its existence alone is not live-state proof.

Size: 537 bytes | 16 text lines

Top-level fields: chain_id, sender, recipient, role, amount_test_POL, nonce, transaction_hash, state, created_utc, block, gas_used, recipient_test_POL, sender_remaining_test_POL, confirmations

UI_SEARCH_SCORING_VERIFICATION.md

Project documentation for UI SEARCH SCORING VERIFICATION. Treat dated plans and old status notes as historical unless confirmed by current code/deployment.

Size: 4,330 bytes | 67 text lines

Document sections: UI, search and assessment verification; Delivered; Actual checks; Limitations relevant to the research paper

17. Current local runbook and troubleshooting

These instructions describe the existing local setup; they do not reinstall, reset, redeploy or alter saved data. Start Docker Desktop if your PostgreSQL service is containerized, and preserve its existing volume. Do not replace a database or regenerate the master key to fix a login/startup error.

```
Backend (from backend/):
  .\venv\Scripts\python.exe -m uvicorn main:app --host 127.0.0.1 --port 8000

Frontend (from frontend/):
  npm run dev
  # For a production preview, build first, then npm run start

Open: http://localhost:3000
Network in MetaMask: Polygon Amoy, chain ID 80002
```

Ensure the existing backend environment references the intended database, Ollama endpoint/model, Pinata credential, content master key and current deployment. Values are private and are not copied here. Provider and recorder workers should remain disabled for the wallet-only demonstration unless separately needed and configured. A successful health response may check configuration/presence rather than execute every external operation.

Symptom	Likely distinction / next check
Insufficient USDC with POL present	Read MockUSDC balance, not native POL or its displayed dollar estimate
Gas tip below minimum	Use shared Amoy fee preparation; do not reduce below required RPC floor
Gas limit above RPC cap	Estimate the actual call, avoid enormous fixed gas overrides
PriceTooLow on old registry	Confirm current successor address; changing local source cannot update old bytecode
Upload exists but mint failed	Resume the saved draft; check pending/completed mint before retry
Duplicate search entries / missing View	Published-only token links and content/token deduplication; do not delete source data blindly
Cancel listing fails	Use the original market for old listings; recheck owner/active state and gas
Purchase succeeded but no download	Verify confirmed entitlement, linked CID row, wrapped key and gateway availability
Ollama / embedding failure	Check local model availability; do not fabricate a fallback quality score
History incomplete	Check indexer coverage/checkpoints; partial history is not zero historical activity

18. Scope, reproducibility and final handoff

This snapshot documents every file returned by the repository-aware inventory after the stated exclusions. It is not a literal dump of every dependency, generated artifact, secret or temporary file on disk. The atlas includes file roles and main code symbols; selected core implementations are reproduced in Chapter 15. For exact full source, use the matching working-tree snapshot, not a reformatted PDF.

File category	Treatment
Project source/config/tests/docs/evidence	Per-file atlas, symbols, routes and source references
.env / private keys / credential files	Contents excluded; referenced variable names only
node_modules / Python virtual environments	Third-party dependency trees summarized by manifests/lockfiles
.next / artifacts / cache / bytecode build output	Generated derivatives; regenerate with the pinned build toolchain
tmp/	Transient investigation/rendering scripts and scratch files; not canonical app source
output/	Generated reports and plots; evidence sources referenced where used, handbook itself excluded from recursive inventory
Git internals / logs / database dumps	Not reproduced; may contain history, secrets or user data

The current working branch is codex/dkm-payment-persistence; the handbook describes uncommitted working-tree changes as well as previously saved code. Before publishing research artifacts, freeze a commit, record lockfiles and exact environment, and remove secrets/user data from the release package. No code was pushed or deployed to create this document.

Primary internal evidence

- PROJECT_CONTEXT.md: latest implementation/deployment and test history.

- TEST_EVIDENCE.md and acceptance_results.json: prior test outcomes and limitations.

- contracts/cents-deployment.json: current deployment receipts and preserved legacy state.

- research_results/vqc_embeddings.json and qaoa_pricing.json: experimental results; companion reports explain protocols.

- restore_drill_evidence.json and SLITHER_AUDIT.json: bounded restore/static-analysis evidence.

- output/research/ui_search_scoring_checks.json: retained retrieval/scoring sanity evidence.

- Confirmed test-funding transaction shown in Chapter 9: receipt and balance verified during the prior funding task.

No new all-suite test run, real-money payment, complete purchase/download, independent security audit or quantum-hardware experiment was performed for this handbook. The last confirmed live step is test-token funding. The remaining demonstration must show the actual purchase entitlement and successful authorized download, with an unrelated wallet denied access.